

Primitive Roots

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1 The Order of a Number modulo m

Let a and m be relatively prime integers, with $m \geq 1$, and consider all the positive powers of a :

$$a, a^2, a^3, \dots$$

We know, from the *Euler-Fermat Theorem*, that $a^{\varphi(m)} \equiv 1 \pmod{m}$. However, there may be an earlier power a^f such that $a^f \equiv 1 \pmod{m}$. We are interested in the smallest f with this property.

Definition 1. The smallest positive integer f such that $a^f \equiv 1 \pmod{m}$ is called the **order** of a modulo m , and is denoted by $f = \text{ord}_m a$. If $\text{ord}_m a = \varphi(m)$ then a is called a **primitive root** mod m (or sometimes we also say that a is a primitive root of m).

The *Euler-Fermat Theorem* tells us that $\text{ord}_m a \leq \varphi(m)$. The next theorem shows that $\text{ord}_m a$ divides $\varphi(m)$.

Theorem 1. Let $m \geq 1$, $\gcd(a, m) = 1$ and $f = \text{ord}_m a$. Then

- (a) $a^k \equiv a^h \pmod{m}$ if and only if $k \equiv h \pmod{f}$.
- (b) $a^k \equiv 1 \pmod{m}$ if and only if $k \equiv 0 \pmod{f}$. In particular, $f \mid \varphi(m)$.
- (c) The numbers $1, a, a^2, \dots, a^{f-1}$ are incongruent mod m .

PROOF

Parts (b) and (c) follow at once from (a), so we need only prove (a). If $a^k \equiv a^h \pmod{m}$ then $a^{k-h} \equiv 1 \pmod{m}$. Write

$$k - h = qf + r \quad \text{where } 0 \leq r < f.$$

So $1 \equiv a^{k-h} \equiv a^{qf+r} \equiv a^r \pmod{m}$ which implies that $r = 0$ (by the minimality of f) and so $k \equiv h \pmod{f}$.

Conversely, if $k \equiv h \pmod{f}$ then $k - h = qf$ for some integer q and so $a^{k-h} = a^{qf} = (a^f)^q \equiv 1^q = 1 \pmod{m}$. Hence $a^k \equiv a^h \pmod{m}$. $\square$

2 Primitive Roots and Reduced Residue Systems

Theorem 2. Let $\gcd(a, m) = 1$. Then a is primitive root mod m if and only if the numbers

$$(1) \quad a, a^2, \dots, a^{\varphi(m)}$$

form a reduced residue system mod m . In particular, $a, a^2, \dots, a^{p-1}$ are congruent mod p , in some order, to the numbers $1, 2, \dots, p-1$ if and only if a is a primitive root mod p .

PROOF

If a is a primitive root the numbers in (1) are incongruent mod m . Since there are $\varphi(m)$ such numbers they form a reduced residue system mod m .

Conversely, if the numbers in (1) form a reduced residue system, then $a^{\varphi(m)} \equiv 1 \pmod{m}$ but no smaller power is congruent to 1, so a is a primitive root. $\square$

3 Existence and Nonexistence of Primitive Roots

The Nonexistence of Primitive Roots mod 2^α for $\alpha \geq 3$

Theorem 3. *Let x be an odd integer. If $\alpha \geq 3$, then*

$$(2) \quad x^{\frac{\varphi(2^\alpha)}{2}} \equiv 1 \pmod{2^\alpha},$$

so there are no primitive roots mod 2^α .

PROOF

If $\alpha = 3$, congruence (2) states that $x^2 \equiv 1 \pmod{8}$ for x odd. This is easily verified by testing $x = 1, 3, 5, 7$ or by noting that

$$(2k+1)^2 = 4k(k+1) + 1$$

and observing that $k(k+1)$ is even.

Now we prove the theorem by induction on α . We assume (2) holds for α and prove that it holds for $\alpha + 1$. The induction hypothesis is that

$$x^{\frac{\varphi(2^\alpha)}{2}} = 1 + 2^\alpha t,$$

where t is an integer. Squaring both sides we obtain

$$x^{\varphi(2^\alpha)} = 1 + 2^{\alpha+1}t + 2^{2\alpha}t^2 \equiv 1 \pmod{2^{\alpha+1}}$$

because $2\alpha \geq \alpha + 1$. This completes the proof since $\varphi(2^\alpha) = 2^{\alpha-1} = \frac{\varphi(2^{\alpha+1})}{2}$. □

The Existence of Primitive Roots mod p for odd primes p

Lemma 1. *Given $\gcd(a, m) = 1$, let $f = \text{ord}_m a$. Then*

$$\text{ord}_m(a^k) = \frac{\text{ord}_m a}{\gcd(k, f)}.$$

In particular, $\text{ord}_m(a^k) = \text{ord}_m a$ if and only if $\gcd(k, f) = 1$.

PROOF

The order of a^k is the smallest positive integer x such that $a^{kx} \equiv 1 \pmod{m}$. This is also the smallest positive integer x such that $kx \equiv 0 \pmod{f}$. But this latter congruence is equivalent to the congruence

$$x \equiv 0 \pmod{\frac{f}{d}}$$

where $d = \gcd(k, f)$. The smallest positive solution of this congruence is $\frac{f}{d}$, and so $\text{ord}_m(a^k) = \frac{f}{d}$ as asserted. □

Theorem 4. *Let p be an odd prime and let d be any positive divisor of $p-1$. Then in every reduced residue system mod p there are exactly $\varphi(d)$ numbers a such that*

$$\text{ord}_p a = d.$$

In particular, when $d = \varphi(p) = p-1$ there are exactly $\varphi(p-1)$ primitive roots mod p .

PROOF

The numbers $1, 2, \dots, p-1$ are distributed into disjoint sets $A(d)$, each set corresponding to a divisor d of $p-1$. Here we define

$$A(d) = \{x : 1 \leq x \leq p-1 \text{ and } \text{ord}_p x = d\}.$$

Let $f(d)$ be the number of elements of $A(d)$. Then $f(d) \geq 0$ for each d . The goal is to prove that $f(d) = \varphi(d)$.

Since the sets $A(d)$ are disjoint and since each $x = 1, 2, \dots, p-1$ falls into some $A(d)$, we have

$$\sum_{d|p-1} f(d) = p-1.$$

But we also have

$$\sum_{d|p-1} \varphi(d) = p-1$$

so

$$\sum_{d|p-1} (\varphi(d) - f(d)) = 0.$$

To show each term in this sum is zero it is enough to prove that $f(d) \leq \varphi(d)$. We do this by showing either $f(d) = 0$ or $f(d) = \varphi(d)$; in other words, that $f(d) \neq 0$ implies that $f(d) = \varphi(d)$.

Suppose that $f(d) \neq 0$. Then $A(d) \neq \emptyset$ and so $a \in A(d)$ for some positive integer a . Consequently $\text{ord}_p a = d$. Hence $a^d \equiv 1 \pmod{p}$. But every power of a satisfies the same congruence, so the d numbers $a, a^2, \dots, a^d$ are solutions of the polynomial congruence $x^d - 1 \equiv 0 \pmod{p}$, these solutions being incongruent mod p since $d = \text{ord}_p a$. Moreover note that this polynomial congruence has at most d solutions since the modulus is prime. So the d numbers $a, a^2, \dots, a^d$ must be all the solutions of the said polynomial congruence. Hence each number in $A(d)$ must be of the form a^k for some $k = 1, 2, \dots, d$. When is $\text{ord}_p(a^k) = d$? According to Lemma 1, this occurs if and only if $\gcd(k, d) = 1$. In other words, among the d numbers $a, a^2, \dots, a^d$ there are $\varphi(d)$ which have order d mod p . Thus we have shown that $f(d) = \varphi(d)$ if $f(d) \neq 0$. $\square$

Theorem 5. *Let a be a primitive root mod m . Then a^k is a primitive root of m if and only if $\gcd(k, \varphi(m)) = 1$.*

PROOF

Since $\text{ord}_m a = \varphi(m)$, it follows from Lemma 1 that

$$\text{ord}_m(a^k) = \frac{\varphi(m)}{\gcd(k, \varphi(m))}.$$

So a^k will be a primitive root mod m (that is, $\text{ord}_m(a^k) = \varphi(m)$) if and only if $\gcd(k, \varphi(m)) = 1$. $\square$

The Existence of Primitive Roots mod p^α

We turn next to the case $m = p^\alpha$, where p is an odd prime and $\alpha \geq 2$. In seeking primitive roots mod p^α it is natural to consider as candidates the primitive roots mod p . Let g be such a primitive root and let us ask whether g might also be a primitive root mod p^2 . Now $g^{p-1} \equiv 1 \pmod{p}$ and since $\varphi(p^2) = p(p-1) > p-1$, this g will certainly not be a primitive root mod p^2 if $g^{p-1} \equiv 1 \pmod{p^2}$. Therefore the relation

$$g^{p-1} \not\equiv 1 \pmod{p^2}$$

is a necessary condition for a primitive root $g \pmod{p}$ to be also a primitive root mod p^2 . Remarkably enough, this condition is also sufficient for g to be a primitive root mod p^2 and, more generally, mod p^α for all powers $\alpha \geq 2$.

Theorem 6. *Let p be an odd prime.*

- (a) *If g is a primitive root mod p then g is also a primitive root mod p^α for all $\alpha \geq 1$ if and only if*

$$(3) \quad g^{p-1} \not\equiv 1 \pmod{p^2}.$$

- (b) *There is at least one primitive root $g \pmod{p}$ which satisfies (3), hence there exists at least one primitive root mod p^α if $\alpha \geq 2$.*

PROOF

We prove (b) first. Let g be a primitive root mod p . If $g^{p-1} \not\equiv 1 \pmod{p^2}$ there is nothing to prove. However, if $g^{p-1} \equiv 1 \pmod{p^2}$ we can show that $g_1 = g + p$, which is another primitive root mod p , satisfies the condition

$$g_1^{p-1} \not\equiv 1 \pmod{p^2}.$$

In fact, we have

$$\begin{aligned} g_1^{p-1} &= (g + p)^{p-1} \\ &= g^{p-1} + (p-1)g^{p-2}p + tp^2 \\ &\equiv g^{p-1} + (p^2 - p)g^{p-2} \pmod{p^2} \\ &\equiv 1 - pg^{p-2} \pmod{p^2}. \end{aligned}$$

But we cannot have $pg^{p-2} \equiv 0 \pmod{p^2}$ for this would imply $g^{p-2} \equiv 0 \pmod{p}$, contradicting the fact that g is a primitive root mod p . Hence $g_1^{p-1} \not\equiv 1 \pmod{p^2}$, so (b) is proved.

Now we prove (a). Let g be a primitive root mod p . If this g is a primitive root mod p^α for all $\alpha \geq 1$ then, in particular, it is a primitive root mod p^2 and as we have already noted, this implies (3).

Now we prove the converse statement. Suppose that g is a primitive root mod p which satisfies (3). We must show that g is also a primitive root mod p^α for all $\alpha \geq 2$. Let t be the order of $g \pmod{p^\alpha}$. We wish to show that $t = \varphi(p^\alpha)$. Since $g^t \equiv 1 \pmod{p^\alpha}$ we also have $g^t \equiv 1 \pmod{p}$ so $\varphi(p) \mid t$ and we can write

$$t = q\varphi(p).$$

Now $t \mid \varphi(p^\alpha)$ so $q\varphi(p) \mid \varphi(p^\alpha)$. But $\varphi(p^\alpha) = p^{\alpha-1}(p-1)$ hence

$$q(p-1) \mid p^{\alpha-1}(p-1)$$

which means $q \mid p^{\alpha-1}$. Therefore $q = p^\beta$ where $\beta \leq \alpha - 1$. So

$$t = p^\beta(p-1).$$

If we prove that $\beta = \alpha - 1$ then $t = \varphi(p^\alpha)$ and the proof will be complete. Suppose, on the contrary, that $\beta < \alpha - 1$. Then $\beta \leq \alpha - 2$ and we have

$$t = p^\beta(p-1) \mid p^{\alpha-2}(p-1) = \varphi(p^{\alpha-1}).$$

Thus, since $\varphi(p^{\alpha-1})$ is a multiple of t , this implies

$$(4) \quad g^{\varphi(p^{\alpha-1})} \equiv 1 \pmod{p^\alpha}.$$

But now we make use of the following Lemma which shows that (4) is a contradiction. $\square$

Lemma 2. *Let g be a primitive root mod p such that*

$$(5) \quad g^{p-1} \not\equiv 1 \pmod{p^2}.$$

Then, for every $\alpha \geq 2$,

$$(6) \quad g^{\varphi(p^{\alpha-1})} \not\equiv 1 \pmod{p^\alpha}.$$

PROOF

We use induction on α . For $\alpha = 2$, the relation (6) reduces to (5). Suppose then, that (6) holds for α . By the *Euler-Fermat Theorem*,

$$g^{\varphi(p^{\alpha-1})} \equiv 1 \pmod{p^{\alpha-1}}$$

so

$$g^{\varphi(p^{\alpha-1})} = 1 + kp^{\alpha-1}$$

where $p \nmid k$ because of (5). Raising both sides of this relation to the p th power we find

$$g^{\varphi(p^\alpha)} = (1 + kp^{\alpha-1})^p = 1 + kp^\alpha + k^2 \frac{p(p-1)}{2} p^{2(\alpha-1)} + rp^{3(\alpha-1)}.$$

Now $2\alpha - 1 \geq \alpha + 1$ and $3\alpha - 3 \geq \alpha + 1$ since $\alpha \geq 2$. Hence, the last equation gives us the congruence

$$g^{\varphi(p^\alpha)} \equiv 1 + kp^\alpha \pmod{p^{\alpha+1}}$$

where $p \nmid k$. In other words, $g^{\varphi(p^\alpha)} \not\equiv 1 \pmod{p^{\alpha+1}}$ so (5) holds for $\alpha + 1$ if it holds for α . $\square$

The Existence of Primitive Roots mod $2p^\alpha$

Theorem 7. *If p is an odd prime and $\alpha \geq 1$ there exist odd primitive roots $g \bmod p^\alpha$. Each such g is also a primitive root mod $2p^\alpha$.*

PROOF

If g is a primitive root mod p^α so is $g + p^\alpha$. But one of g or $g + p^\alpha$ is odd so odd primitive roots mod p^α always exist. Let g be an odd primitive root mod p^α and let f be the order of g mod $2p^\alpha$. We wish to show that $f = \varphi(2p^\alpha)$. Now $f \mid \varphi(2p^\alpha)$, and

$$\varphi(2p^\alpha) = \varphi(2)\varphi(p^\alpha) = \varphi(p^\alpha)$$

so $f \mid \varphi(p^\alpha)$. On the other hand, $g^f \equiv 1 \pmod{2p^\alpha}$ so $g^f \equiv 1 \pmod{p^\alpha}$, hence $\varphi(p^\alpha) \mid f$ since g is a primitive root mod p^α . Therefore $f = \varphi(p^\alpha) = \varphi(2p^\alpha)$, so g is a primitive root mod $2p^\alpha$. $\square$

Nonexistence of Primitive Roots in the Remaining Cases

Theorem 8. *Given $m \geq 1$ where m is not of the form $m = 1, 2, 4, p^\alpha$, or $2p^\alpha$, where p is an odd prime. Then for any a with $\gcd(a, m) = 1$,*

$$a^{\frac{\varphi(m)}{2}} \equiv 1 \pmod{m},$$

so there are no primitive roots mod m .

PROOF

We have already shown that there are no primitive roots mod 2^α if $\alpha \geq 3$. Therefore we can suppose that m has the factorization

$$m = 2^\alpha p_1^{\alpha_1} \cdot \dots \cdot p_s^{\alpha_s}$$

where the p_i are odd primes, $s \geq 1$, and $\alpha \geq 0$. Since m is not of the form $1, 2, 4, p^\alpha$, or $2p^\alpha$ we have $\alpha \geq 2$ if $s = 1$ and $s \geq 2$ if $\alpha = 0$ or 1 . Note that

$$\varphi(m) = \varphi(2^\alpha)\varphi(p_1^{\alpha_1}) \cdot \dots \cdot \varphi(p_s^{\alpha_s}).$$

Now let a be any integer relatively prime to m . We wish to prove that

$$a^{\frac{\varphi(m)}{2}} \equiv 1 \pmod{m}.$$

Let g be a primitive root mod $p_1^{\alpha_1}$ and choose k so that

$$a \equiv g^k \pmod{p_1^{\alpha_1}}.$$

Then

$$(7) \quad a^{\frac{\varphi(m)}{2}} \equiv g^{\frac{k\varphi(m)}{2}} \equiv g^{t\varphi(p_1^{\alpha_1})} \pmod{p_1^{\alpha_1}}$$

where

$$t = \frac{k\varphi(2^\alpha)\varphi(p_2^{\alpha_2}) \cdot \dots \cdot \varphi(p_s^{\alpha_s})}{2}.$$

We will show that t is an integer. If $\alpha \geq 2$ the factor $\varphi(2^\alpha)$ is even and hence t is an integer. If $\alpha = 0$ or 1 then $s \geq 2$ and the factor $\varphi(p_2^{\alpha_2})$ is even, so t is an integer in this case as well. Hence congruence (7) gives us

$$a^{\frac{\varphi(m)}{2}} \equiv 1 \pmod{p_1^{\alpha_1}}.$$

In the same way we find

$$(8) \quad a^{\frac{\varphi(m)}{2}} \equiv 1 \pmod{p_i^{\alpha_i}}$$

for each $i = 1, 2, \dots, s$. Now we show that this congruence holds mod 2^α . If $\alpha \geq 3$ the condition $\gcd(a, m) = 1$ requires a to be odd and we may apply Theorem 3 to write

$$a^{\frac{\varphi(2^\alpha)}{2}} \equiv 1 \pmod{2^\alpha}.$$

Since $\varphi(2^\alpha) \mid \varphi(m)$ this gives us

$$(9) \quad a^{\frac{\varphi(m)}{2}} \equiv 1 \pmod{2^\alpha}$$

for $\alpha \geq 3$.

If $\alpha \leq 2$ we have

$$(10) \quad a^{\frac{\varphi(2^\alpha)}{2}} \equiv 1 \pmod{2^\alpha}.$$

But $s \geq 1$ so $\varphi(m) = \varphi(2^\alpha)\varphi(p_1^{\alpha_1})\dots\varphi(p_s^{\alpha_s}) = 2r\varphi(2^\alpha)$ where r is an integer. Hence $\varphi(2^\alpha) \mid \varphi(m)/2$ and (10) implies (9) for $\alpha \leq 2$. Hence (9) holds for all α .

Multiplying together the congruences (8) and (10) we get

$$a^{\frac{\varphi(m)}{2}} \equiv 1 \pmod{m},$$

and this shows that a cannot be a primitive root mod m . □

4 Primitive Roots and Quadratic Residues

Theorem 9. *Let g be a primitive root mod p , where p is an odd prime. Then the even powers*

$$g^2, g^4, \dots, g^{p-1}$$

are quadratic residues mod p , and the odd powers

$$g, g^3, \dots, g^{p-2}$$

are quadratic nonresidues mod p .

PROOF

If n is even, say $n = 2m$, then $g^n = (g^m)^2$ so

$$g^n \equiv x^2 \pmod{p}, \quad \text{where } x = g^m.$$

Hence g^n is a quadratic residue mod p . But there are $\frac{p-1}{2}$ distinct even powers $g^2, g^4, \dots, g^{p-1}$ mod p and the same number of quadratic residues mod p . Therefore the even powers are the quadratic residues and the odd powers are the nonresidues. $\square$

5 The Number of Primitive Roots

Theorem 10. *If m has a primitive root g then m has exactly $\varphi(\varphi(m))$ incongruent primitive roots and they are given by the numbers in the set*

$$S = \{g^n : 1 \leq n \leq \varphi(m) \text{ and } \gcd(n, \varphi(m)) = 1\}.$$

PROOF

We have $\text{ord}_m g = \varphi(m)$ and Lemma 1 shows that $\text{ord}_m(g^k) = \text{ord}_m g$ if and only if

$$\gcd(n, \varphi(m)) = 1.$$

Therefore each element of S is a primitive root mod m .

Conversely, if a is a primitive root mod m , then $a \equiv g^k \pmod{m}$ for some $k = 1, \dots, \varphi(m)$. Hence $\text{ord}_m(g^k) = \text{ord}_m a = \varphi(m)$, and Lemma 1 implies $\gcd(n, \varphi(m)) = 1$. Therefore every primitive root is a member of S . Since S contains $\varphi(\varphi(m))$ incongruent members mod m the proof is complete. $\square$

6 Miscellaneous Examples

Example 1. Let $\gcd(a, m) = 1$. If a has order $m - 1 \pmod{m}$, then m is prime.

PROOF

Note that the order of a cannot be larger than $\varphi(m)$, that is, $m - 1 \leq \varphi(m)$. Since $\varphi(m) \leq m - 1$, it follows that $\varphi(m) = m - 1$. Therefore m is prime. $\square$

Example 2. Let $n > 1$. Prove that $\varphi(2^n - 1)$ is divisible by n .

PROOF

Note that $2^n \equiv 1 \pmod{2^n - 1}$. Moreover observe that if $1 \leq k < n$, then $2 \leq 2^k < 2^n$. So $2^k \not\equiv 1 \pmod{2^n - 1}$. Hence it follows from the definition of the order that 2 has order $n \pmod{2^n - 1}$. Therefore, by Theorem 1, $\varphi(2^n - 1)$ is divisible by n . $\square$

Example 3. Suppose p is an odd prime and $a \not\equiv \pm 1 \pmod{p}$. Is it true that if $a^r \not\equiv \pm 1 \pmod{p}$ for some positive integer $r < p - 1$ then $r \mid p - 1$?

PROOF

No. Counterexample: let $p = 13$. Then $5^4 \not\equiv \pm 1 \pmod{13}$. So $5^8 \not\equiv \pm 1 \pmod{13}$ but $8 \nmid 12 = 13 - 1$. (Note that Theorem 1 asserts only that the order of a divides $p - 1$.) $\square$

Example 4. Let p be a prime with $\gcd(p, 10) = 1$ and let n be a positive integer with $n < p$. If d is the order of $10 \pmod{p}$, then the length of the period of the decimal expansion of n/p is d .

PROOF

Let m be the length of the period and let $\frac{n}{p} = 0.\overline{a_1 a_2 \dots a_m}$. Then

$$\frac{10^m n}{p} = a_1 a_2 \dots a_m \cdot \overline{a_1 a_2 \dots a_m},$$

and so

$$\frac{(10^m - 1)n}{p} = a_1 a_2 \dots a_m$$

is an integer. Since n and p are relatively prime, p must divide $10^m - 1$, so $d \mid m$. Conversely, $p \mid 10^d - 1$ and so

$$\frac{(10^d - 1)n}{p}$$

is an integer with at most d digits. If we divide this integer by $10^d - 1$, then we obtain a rational number whose decimal expansion has period at most d . Therefore $m = d$. $\square$

Example 5. Let p be a prime with $\gcd(p, 10) = 1$ and let n be a positive integer with $n < p$. If d is the order of $10 \bmod p$ and d is even, then the period of the decimal expansion of n/p can be divided into two halves, whose sum is $10^{\frac{d}{2}} - 1$. For example, $1/7 = 0.\overline{142857}$, so $d = 6$, and $142 + 857 = 999 = 10^3 - 1$.

PROOF

Let $d = 2k$. Then $n/p = 0.\overline{a_1 \dots a_k a_{k+1} \dots a_{2k}}$. Now $p \mid 10^d - 1 = 10^{2k} - 1 = (10^k - 1)(10^k + 1)$. However $p \nmid 10^k - 1$ (since the order of 10 is $2k$), so $p \mid 10^k + 1$. Hence

$$\frac{10^k n}{p} = a_1 \dots a_k \cdot \overline{a_{k+1} \dots a_{2k}},$$

which implies that

$$\frac{(10^k + 1)n}{p} = a_1 a_2 \dots a_k + 0.\overline{a_1 \dots a_k} + 0.\overline{a_{k+1} \dots a_{2k}}$$

is an integer. This can occur if and only if $a_1 \dots a_k + a_{k+1} \dots a_{2k}$ is a number consisting only of 9's, and hence, equal to $10^k - 1$. $\square$

Example 6. Let $\gcd(a, m) = \gcd(b, m) = 1$. If $\gcd(\text{ord}_m a, \text{ord}_m b) = 1$, then $\text{ord}_m(ab) = \text{ord}_m a \cdot \text{ord}_m b$.

PROOF

Let $h = \text{ord}_m a$, $k = \text{ord}_m b$ and $r = \text{ord}_m(ab)$. Clearly, $(ab)^{hk} = (a^h)^k (b^k)^h \equiv 1 \pmod{m}$. Then $r \mid hk$ by Theorem 1. Also, $b^{rh} \equiv (a^h)^r b^{rh} = (ab)^{rh} \equiv 1 \pmod{m}$. Hence $k \mid rh$. Since $\gcd(h, k) = 1$, it follows that $k \mid r$. In a similar way, we can show that $h \mid r$, and therefore $hk \mid r$ since $\gcd(h, k) = 1$. Thus $r = hk$. $\square$

Example 7. Let $\gcd(g, m) = 1$. Then g is a primitive root mod m if and only if $g^{\frac{\varphi(m)}{q}} \not\equiv 1 \pmod{m}$ for every prime factor of $\varphi(m)$.

PROOF

If g is a primitive root mod m , then $g^{\frac{\varphi(m)}{q}} \not\equiv 1 \pmod{m}$ for any prime q , for if $g^{\frac{\varphi(m)}{q}} \equiv 1 \pmod{m}$, then the order of g is less than $\varphi(m)$. Conversely, suppose that $\gcd(g, m) = 1$ and that g is not a primitive root mod m . Then g has order d for some $d < \varphi(m)$, and $d \mid \varphi(m)$ by Theorem 1. Now let $\varphi(m) = dk$ and let q be a prime factor of k . Then $\frac{\varphi(m)}{q}$ is a multiple of d , and since $g^d \equiv 1 \pmod{m}$, it follows that $g^{\frac{\varphi(m)}{q}} \equiv 1 \pmod{m}$. $\square$

Example 8 (Lucas). Let m be an integer greater than 1. Suppose there is an integer a such that $a^{m-1} \equiv 1 \pmod{m}$ and $a^{\frac{m-1}{q}} \not\equiv 1 \pmod{m}$ for every prime factor of $m-1$. Then m is prime.

PROOF

Let d be the order of $a \bmod m$. If $a^{m-1} \equiv 1 \pmod{m}$, then by Theorem 1, $d \mid m-1$. If $d < m-1$, then $m-1 = dk$ for some positive integer $k \geq 2$. Now let q be a prime divisor of k . Hence $\frac{m-1}{q}$ is a multiple of d , and since $a^d \equiv 1 \pmod{m}$, it follows that $a^{\frac{m-1}{q}} \equiv 1 \pmod{m}$, a contradiction. So $d = m-1$ and consequently a has order $m-1$. Since $m-1 = \text{ord}_m(a) \leq \varphi(m) \leq m-1$, it follows that $\varphi(m) = m-1$. Therefore m is prime. $\square$

Example 9. Prove that 3 is a primitive root of 17.

PROOF

We will show that 3 has order 16 mod 17. Now the order of 3 divides $\varphi(17) = 16$, so, by Example 7, it is enough to check that $3^8 \not\equiv 1 \pmod{17}$. This can be done by direct computation. Alternatively, observe that 3 is a quadratic nonresidue of 17, since, by the *Quadratic Reciprocity Law*,

$$\left(\frac{3}{17}\right) = (-1)^{\frac{(3-1)(17-1)}{4}} \left(\frac{17}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

Thus, by *Euler's Criterion*, $3^8 \equiv -1 \pmod{17}$. □

Example 10. Show that 2 is a primitive root of the prime 101.

PROOF

Let d be the order of 2 mod 101. Then $d \mid \varphi(101) = 100$. So it is enough to prove that $2^{50} \not\equiv 1 \pmod{101}$ and $2^{20} \not\equiv 1 \pmod{101}$. Now

$$2^{50} = 1024^5 \equiv 14^5 = 196 \cdot 196 \cdot 14 \equiv (-6) \cdot (-6) \cdot 14 \equiv -1 \pmod{101},$$

and

$$2^{20} = 1024^2 \equiv 14^2 \equiv -6 \pmod{101}.$$

Therefore $d = 100$ and so 2 is a primitive root mod 101. □

Example 11. Find all of the primitive roots of 29 between 1 and 28.

SOLUTION

First note that 2 is a primitive root of 29. To prove this, so by Example 7, it is enough to show that $2^{14} \not\equiv 1 \pmod{29}$ and $2^4 \not\equiv 1 \pmod{29}$. The first congruence can be proved by noting that 29 is a prime of the form $8k + 5$ and so 2 is a quadratic nonresidue of 29. By *Euler's Criterion*, $2^{14} \equiv -1 \pmod{29}$. The second congruence is obvious.

By Theorem 5, the primitive roots of 29 are precisely the numbers congruent mod 29 to 2^k where $\gcd(k, 28) = 1$. It follows that the primitive roots of 29 between 1 and 28 are 2, 8, 3, 19, 18, 14, 27, 21, 26, 10, 11, 15. □

Example 12. Prove that if the order of $a \pmod{m}$ is $4r$, then the order of $-a \pmod{m}$ is also $4r$.

PROOF

Since $(-a)^{4r} = a^{4r} \equiv 1 \pmod{m}$, Theorem 1 implies that the order of $-a$ divides $4r$. If the order of $-a$ is the even number $2s < 4r$, then $a^{2s} = (-a)^{2s} \equiv 1 \pmod{m}$. This contradicts the fact that $4r$ is the order of $a \pmod{m}$. If the order of $-a$ is the odd number t , then $t \mid 4r$ which in turn implies that $t \mid r$. So $(-a)^r \equiv 1 \pmod{m}$. Consequently $a^{2r} = (-a)^{2r} \equiv 1 \pmod{m}$. This again contradicts the fact that $4r$ is the order of $a \pmod{m}$. □

Example 13. Suppose g is a primitive root of m and $gh \equiv 1 \pmod{m}$. Prove that h is also a primitive root of m .

PROOF

For every positive integer n , $1 \equiv (gh)^n = g^n h^n \pmod{m}$. Thus $g^n \equiv 1 \pmod{m}$ if and only if $h^n \equiv 1 \pmod{m}$. In particular, g and h have the same order. Therefore if one is a primitive root of m , then so is the other. $\square$

Example 14. Show that if $m_1, \dots, m_r$ are relatively prime in pairs and each m_i has a primitive root, then there is an integer g that is simultaneously a primitive root of every m_i .

PROOF

Let g_i be a primitive root of m_i . Then, by the *Chinese Remainder Theorem*, there is an integer g such that $g \equiv g_i \pmod{m_i}$ for every i . Then g is a primitive root of m_i for all i . $\square$

Example 15. Suppose $p > 3$ is prime. Prove that the product of the primitive roots of p between 1 and $p - 1$ is congruent to 1 mod p .

PROOF

If g is a primitive root of p , then $\gcd(g, p) = 1$. So g has an inverse $h \pmod{p}$. By Example 13, h is also a primitive root of p . If g is a primitive root of p , then its inverse h is not congruent to $g \pmod{p}$. Otherwise, we would have $g^2 \equiv 1 \pmod{p}$, giving $g \equiv \pm 1 \pmod{p}$; but neither 1 or -1 is a primitive root of any prime greater than 3. Therefore the primitive roots of p can be arranged in pairs that have a product congruent to 1 mod p , and hence the product of the primitive roots of p is congruent to 1 mod p . $\square$

Example 16. Suppose m has a primitive root, and let $a_1, a_2, \dots, a_{\varphi(m)}$ be a reduced residue system mod m . Prove that $a_1 a_2 \cdots a_{\varphi(m)} \equiv -1 \pmod{m}$. (Note that if m is prime, this is simply *Wilson's Theorem*).

PROOF

The result is trivial for $m = 1$ and $m = 2$, so we may take $m \geq 3$. Let g be a primitive root of m and let $k = \varphi(m)$. By Theorem 2, the elements of a reduced residue system mod m are, in some order, congruent mod m to the numbers $g, g^2, \dots, g^k$. Note that for $m \geq 3$, k is even and so $k + 1$ is odd. Therefore, we have, mod m ,

$$a_1 a_2 \cdots a_{\varphi(m)} \equiv g \cdot g^2 \cdots g^k = g^{\frac{k(k+1)}{2}} = \left(g^{\frac{k}{2}}\right)^{k+1} \equiv (-1)^{k+1} = -1.$$

$\square$

Example 17. Show that 2 is a primitive root of 3^n for all $n \geq 1$.

PROOF

The statement is easily verified for $n = 1$, so assume that the statement holds for $n = k$ for some integer $k \geq 1$, that is, $2^{\varphi(3^k)} \equiv 2^{2 \cdot 3^{k-1}} \equiv 1 \pmod{3^k}$. Now let d be the order of 2 mod 3^{k+1} .

Then $2^d \equiv 1 \pmod{3^{k+1}}$ implies that $2^d \equiv 1 \pmod{3^k}$ and so $2 \cdot 3^{k-1} \mid d$. However $d \mid \varphi(3^{k+1})$ or equivalently, $d \mid 2 \cdot 3^k$. We deduce that d is either $2 \cdot 3^{k-1}$ or $2 \cdot 3^k$.

We need the following claim: $2^{2 \cdot 3^{n-1}} \equiv 1 + 3^n \pmod{3^{n+1}}$ for all $n \geq 1$. Clearly, this is true for $n = 1$. So assume that the statement holds for $n = j$ for some integer $j \geq 1$. Hence, by assumption

$$2^{2 \cdot 3^{j-1}} = 1 + 3^j + 3^{j+1}m$$

for some integer m . Thus

$$2^{2 \cdot 3^j} = 1 + 3^{j+1} + 3^{j+2}M$$

for some integer M . This means that $2^{2 \cdot 3^j} \equiv 1 + 3^{j+1} \pmod{3^{j+2}}$. This proves the claim by induction.

Therefore $2^{2 \cdot 3^{k-1}} \equiv 1 + 3^k \not\equiv 1 \pmod{3^{k+1}}$, so the order of 2 mod 3^{k+1} is $2 \cdot 3^k$, and again, by the induction, the result follows. $\square$

Example 18. Let p be a prime of the form $4k + 1$. If g is a primitive root mod p , then $-g$ is also a primitive root mod p .

PROOF

Since g is a primitive root mod p , it is a quadratic nonresidue of p . Hence, by *Euler's Criterion* and by noting that p is of the form $4k + 1$, we have $g^{2k} \equiv -1 \pmod{p}$. Thus $-g \equiv g^{2k}g = g^{2k+1} \pmod{p}$. But clearly, $\gcd(2k + 1, 4k) = 1$. Thus, by Theorem 5, g^{2k+1} is a primitive root mod p , that is, $-g$ is a primitive root mod p . $\square$

Example 19. Show that -3 is not a primitive root of any prime of the form $4^n + 3$.

PROOF

Let p be a prime of the form $4^n + 3$. Since p is of the form $4k + 3$, it follows that $\left(\frac{-1}{p}\right) = -1$. By the *Quadratic Reciprocity Law*,

$$\left(\frac{3}{p}\right) = (-1)^{\frac{(3-1)(p-1)}{4}} \left(\frac{p}{3}\right) = -\left(\frac{p}{3}\right) = -1,$$

since $p \equiv 1 \pmod{3}$. Thus

$$\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{3}{p}\right) = 1,$$

so -3 is a quadratic residue of p and hence cannot be a primitive root mod p . $\square$

Example 20. Let q and $p = 4q + 1$ be primes. Prove that 2 is a primitive root of p , and show that 3 is a primitive root of p if $q > 3$.

PROOF

Clearly, $q \neq 2$ and q is odd, so p is of the form $8k + 5$. Thus 2 is a quadratic nonresidue of p . Now we will show that 2 has order $p - 1$ mod p . Let $\text{ord}_p 2 = t$. Then $t \mid \varphi(p)$ and so $t \mid 4q$. Therefore $t = 1, 2, 4, q, 2q, 4q$. Since 2 is a quadratic nonresidue of p , by *Euler's Criterion*, $2^{2q} \equiv -1 \pmod{p}$,

and hence t cannot be $1, 2, q, 2q$. Finally, since $p \geq 13$, $p \nmid 2^4 - 1$, and hence $t \neq 4$. Consequently, $t = 4q = p - 1$.

If $q > 3$, then $q \equiv 1 \pmod{3}$, for if $q \equiv 2 \pmod{3}$, then $3 \mid p$. So p must be of the form $12k + 5$. By the *Quadratic Reciprocity Law*,

$$\left(\frac{3}{p}\right) = (-1)^{\frac{(3-1)(p-1)}{4}} \left(\frac{p}{3}\right) = \left(\frac{12k+5}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

Hence 3 is a quadratic nonresidue of p . Now let $\text{ord}_p(3) = t$. Then $t \mid \varphi(p)$ and so $t \mid 4q$. Therefore $t = 1, 2, 4, q, 2q, 4q$. Since 3 is a quadratic nonresidue of p , by *Euler's Criterion*, $3^{2q} \equiv -1 \pmod{p}$, and hence t cannot be $1, 2, q, 2q$. Finally, note that $p \nmid 3^4 - 1$, and hence $t \neq 4$. Consequently, $t = 4q = p - 1$. $\square$

Example 21. Suppose that $n = 3^k$ for $k \geq 1$. Prove that $3n \mid 2^n + 1$.

PROOF

Note that 2 is a primitive root of 3^2 and hence of 3^{k+1} , by Theorem 6. Thus 2 has order $\varphi(3^{k+1}) = 2 \cdot 3^k = 2n \pmod{3^{k+1}}$. In particular, it follows that $2^{2n} \equiv 1 \pmod{3n}$. Therefore 2^n is a solution of the congruence $x^2 \equiv 1 \pmod{3n}$. We cannot have $2^n \equiv 1 \pmod{3n}$, for this would mean that $\text{ord}_{3n} 2 \leq n$. Thus $2^n \equiv -1 \pmod{3n}$, that is, $3n \mid 2^n + 1$. $\square$

Example 22. Let p be an odd prime, and let q and r be primes such that $p \mid q^r + 1$. Prove that either $2r \mid p - 1$ or $p \mid q^2 - 1$.

PROOF

Let $d = \text{ord}_p q$. Since $p \mid q^r + 1$ and $p > 2$, we have $q^r \equiv -1 \not\equiv 1 \pmod{p}$, and so $q^{2r} \equiv (-1)^2 \equiv 1 \pmod{p}$. From these congruences, $d \mid 2r$ but $d \nmid r$. Since r is prime, the only possibilities are $d = 2$ and $d = 2r$. If $d = 2r$, then $2r \mid p - 1$ because $d \mid p - 1$, by Theorem 1. If $d = 2$, then $q^2 \equiv 1 \pmod{p}$ and so $p \mid q^2 - 1$. $\square$

Example 23. Let $a > 1$ and n be given positive integers. If p is a prime divisor of $a^{2^n} + 1$, prove that $p - 1$ is divisible by 2^{n+1} .

PROOF

From the congruence $a^{2^n} \equiv -1 \pmod{p}$, we have $a^{2^{n+1}} = (a^{2^n})^2 \equiv 1 \pmod{p}$. By Theorem 1, $\text{ord}_p a \mid 2^{n+1}$. Since $a^{2^n} \equiv -1 \pmod{p}$, $\text{ord}_p a = 2^{n+1}$. Clearly, $\gcd(a, p) = 1$. By *Fermat's Little Theorem*, we have $a^{p-1} \equiv 1 \pmod{p}$. By Theorem 1, we conclude that $2^{n+1} \mid p - 1$. $\square$

Example 24. Let q be an odd prime and suppose that $p = 2q + 1$ is prime. Then there exist q quadratic nonresidues and $q - 1$ primitive roots of p . Thus the primitive roots of p are just the quadratic nonresidues of p with one exception. What is the exception?

PROOF

There are $\frac{p-1}{2} = q$ quadratic nonresidues of p and $\varphi(p-1) = \varphi(2q) = q-1$ primitive roots of p . Since every primitive root of p is a quadratic nonresidue of p , exactly one quadratic nonresidue of p fails to be a primitive root of p . Since q is odd, it follows that q is of the form $2k + 1$. Thus $p = 2q + 1 = 4k + 3$. So -1 is a quadratic nonresidue of p . Now clearly, -1 has order 2, and so -1 is not a primitive root of p . $\square$

Example 25. Let p be an odd prime. Then every quadratic nonresidue of p is a primitive root of p if and only if p is of the form $2^k + 1$.

PROOF

There are $\frac{p-1}{2}$ quadratic nonresidues of p and $\varphi(p-1)$ primitive roots of p . Since any primitive root is a quadratic nonresidue, it follows that every quadratic nonresidue is a primitive root if and only if $\varphi(p-1) = \frac{p-1}{2}$. But $\varphi(n) = \frac{n}{2}$ if and only if n is of the form 2^k . The result follows. $\square$

Example 26. Let p be an odd prime. Then there are exactly $\varphi(p-1)$ primitive roots of p that are incongruent mod p^2 and that are not primitive roots of p^2 .

PROOF

If g is a primitive root of p , then there are p primitive roots of p that are congruent to $g \pmod{p}$ but incongruent mod p^2 , namely, $g, g+p, g+2p, \dots, g+(p-1)p$. Since p has $\varphi(p-1)$ primitive roots, there are $p\varphi(p-1)$ primitive roots of p that are incongruent mod p^2 . But there are $(p-1)\varphi(p-1)$ primitive roots of p^2 , so the number of primitive roots of p that are incongruent mod p^2 and that are not primitive roots of p^2 is $p\varphi(p-1) - (p-1)\varphi(p-1) = \varphi(p-1)$.

Alternatively, consider $g+tp$, where $0 \leq t \leq p-1$. These are the primitive roots of p that incongruent mod p^2 . By the *Binomial Theorem*, $(g+tp)^{p-1} \equiv g^{p-1} - g^{p-2}tp \pmod{p^2}$. Let $g^{p-1} = 1 + mp$. Then $g^{p-1} \equiv 1 \pmod{p^2}$ if and only if $mp - g^{p-2}tp \equiv 0 \pmod{p^2}$, that is, $mp \equiv g^{p-2}tp \pmod{p}$. Observe that this congruence has a unique solution. Thus $g+tp$ is a primitive root of p for all but one value of t . Since there are $\varphi(p-1)$ primitive roots of p , the result follows. $\square$

Example 27. Let p be a prime number. Find the least positive integer ℓ such that

$$(*) \quad (n + \ell)^{n+\ell} \equiv n^n \pmod{p}$$

for all positive integers n .

PROOF

First we claim that $(n+p(p-1))^{n+p(p-1)} \equiv n^n \pmod{p}$. In fact, if $p \mid n$, then $(n+p(p-1))^{n+p(p-1)} \equiv 0 \equiv n^n \pmod{p}$. If $p \nmid n$, then $n^{p-1} \equiv 1 \pmod{p}$ and so

$$(n + p(p-1))^{n+p(p-1)} \equiv n^{n+p(p-1)} \equiv n^n \pmod{p}.$$

Now let ℓ be the least such positive integer so that $(*)$ holds. Then observe that $p^p \equiv 0 \equiv (p+\ell)^{p+\ell} \pmod{p}$. This forces $p \mid \ell$. Now let $\ell = pr$ for some positive integer r . Let g be a primitive root mod p . Then

$$g^g \equiv (g+pr)^{g+pr} \equiv g^{g+pr} \pmod{p}.$$

Consequently $g^{pr} \equiv 1 \pmod{p}$. Since g is a primitive root mod p , it follows that $p-1 \mid pr$. So $p-1 \mid r$. Therefore $p(p-1) \mid \ell$. Therefore the least such ℓ is $p(p-1)$. $\square$

Example 28 (AIME 2001). How many positive integer multiples of 1001 can be expressed in the form $10^j - 10^i$, where i and j are integers and $0 \leq i < j \leq 99$?

SOLUTION

Because $10^j - 10^i = 10^i(10^{j-i} - 1)$ and $1001 = 7 \cdot 11 \cdot 13$ is relatively prime to 10^i , it is necessary to find i and j such that the $10^{j-i} - 1$ is divisible by the primes 7, 11, and 13. Notice that $10^3 \equiv -1 \pmod{1001}$. It is easy to check that $\text{ord}_{1001} 10 = 6$. By Theorem 1, $10^i(10^{j-i} - 1)$ is divisible by 1001 if and only if $j - i = 6n$ for some positive integer n . Thus it is necessary to count the number of integer solutions to $i + 6n = j$, where $j \leq 99$, $i \geq 0$, and $n > 0$. For each $n = 1, 2, 3, \dots, 15$, there are $100 - 6n$ suitable values of i (and j), so the number of solutions is $94 + 88 + 82 + \dots + 4 = 784$. $\square$

Example 29 (Putnam 1972). Show that if n is an integer greater than 1, then n does not divide $2^n - 1$.

PROOF

Suppose that $n \mid 2^n - 1$ for some $n > 1$. Since $2^n - 1$ is odd, n is odd. Let p be the smallest prime factor of n . Then $p \mid 2^n - 1$ and so $2^n \equiv 1 \pmod{p}$. By *Fermat's Little Theorem*, $2^{p-1} \equiv 1 \pmod{p}$. Then let k be the order of 2 mod p . Hence, by Theorem 1, $k \mid n$ and $k \mid p - 1$. So $k \mid \gcd(n, p - 1)$. Now $d = \gcd(n, p - 1)$ must be 1 since $d \mid n$, $d \leq p - 1$ and p is the smallest prime factor of n . Therefore $k = 1$ and so $2 = 2^k \equiv 1 \pmod{p}$, contradiction. $\square$

Example 30 (Proposed by Romania for IMO 1985). For $k \geq 2$, let $n_1, n_2, \dots, n_k$ be positive integers such that

$$n_2 \mid 2^{n_1} - 1, \quad n_3 \mid 2^{n_2} - 1, \quad \dots, \quad n_k \mid 2^{n_{k-1}} - 1, \quad n_1 \mid 2^{n_k} - 1.$$

Prove that $n_1 = n_2 = \dots = n_k = 1$.

PROOF

Observe that if $n_i = 1$ for some i , then n_{i+1} will equal 1 and the chain effect will cause all of them to be 1. So assume that no n_i is 1. Let p_k be the smallest prime factor of n_k . Then $p_k \mid 2^{n_{k-1}} - 1$. So $2^{n_{k-1}} \equiv 1 \pmod{p_k}$. By *Fermat's Little Theorem*, $2^{p_k-1} \equiv 1 \pmod{p_k}$. Then let m_k be the order of 2 mod p_k . Hence, by Theorem 1, $m_k \mid n_{k-1}$ and $m_k \mid p_k - 1$. In particular, we get $1 < m_k \leq p_k - 1 < p_k$ and so the smallest prime factor p_{k-1} of n_{k-1} is less than p_k . Finally observe that we get the following contradiction: $p_k > p_{k-1} > \dots > p_1 > p_k$ (where this last inequality follows from the last divisibility relation in the given condition). $\square$

Example 31 (IMO 1978). Let m and n be positive integers such that $1 \leq m < n$. In their decimal representations, the last three digits of 1978^m are equal, respectively, to the last three digits of 1978^n . Find m and n such that $m + n$ has its least value.

SOLUTION

Since 1978^n and 1978^m agree in their last three digits, the difference

$$1978^n - 1978^m = 1978^m(1978^{n-m} - 1)$$

is divisible by $10^3 = 2^3 5^3$; and since the second factor above is odd, 2^3 divides the first. Also

$$1978^m = 2^m \cdot 989^m,$$

so $m \geq 3$.

We can write $m + n = (n - m) + 2m$; to minimize this sum, we take $m = 3$ and seek the smallest value of $d = n - m$, such that $1978^d - 1$ is divisible by 5^3 , that is,

$$1978^d \equiv 1 \pmod{125}.$$

This d is nothing but the order of 1978 mod 125. Now observe that $\varphi(125) = 5^3 - 5^2 = 100$ and so, by Theorem 1, $d \mid 100$.

Since $1978^d - 1$ is divisible by 125, it is certainly divisible by 5, so

$$1 \equiv 1978^d \equiv 3^d \pmod{5}.$$

Now the order of 3 mod 5 is 4 and so, by Theorem 1, $4 \mid d$. Since $d \mid 100$, so d must be one of the three values: 4, 20, 100.

We now rule the first two:

$$\begin{aligned} 1978^4 &= (2000 - 22)^4 \equiv (-22)^4 \\ &\equiv (-2)^4 11^4 = (4 \cdot 121)^2 \equiv (4(-4))^2 \\ &\equiv (-16)^2 \equiv 256 \equiv 6 \\ &\not\equiv 1 \pmod{125}. \end{aligned}$$

So $d = n - m \neq 4$. We use this computation to find that

$$1978^{20} = 1978^4 1978^{16} \equiv 6 \cdot 6^4 \equiv 6 \cdot 46 \equiv 26 \not\equiv 1 \pmod{125}.$$

Hence $d = n - m \neq 20$. Therefore $n - m = 100$, $m = 3$, and so $n + m = 106$. □

Example 32 (IMO 1990). Find all positive integers $n > 1$ such that $\frac{2^n + 1}{n^2}$ is an integer.

SOLUTION

Clearly, n must be odd. Now assume that $3^k \parallel n$ (that is, 3^k is the highest power of 3 dividing n). Then $3^{2k} \mid n^2$ and so $2^n \equiv -1 \pmod{3^{2k}}$. Hence $2^{2n} \equiv 1 \pmod{3^{2k}}$. By Example 17, $\varphi(3^{2k}) \mid 2n$. So $3^{2k-1} \mid n$. Thus $2k - 1 \leq k$ which in turn implies that $k \leq 1$, showing that n has at most one factor of 3. We observe that $n = 3$ is a solution.

Suppose that n has a prime factor greater than 3. Let p be the least such a prime. Then $2^n \equiv -1 \pmod{p}$. Now let d be the order of 2 mod p . Since $2^{2n} \equiv 1 \pmod{p}$, it follows, by Theorem 1, that $d \mid 2n$. If d is odd, then $d \mid n$ which means that $2^n \equiv 1 \pmod{p}$, a contradiction. So d must be even, say $d = 2d_1$ for some positive integer d_1 . Then $2d_1 \mid 2n$ implies that $d_1 \mid n$. Also, by *Fermat's Little Theorem*, $d \mid p - 1$. This means that $2d_1 \mid p - 1$. Hence $d_1 \leq \frac{p-1}{2} < p$. Since p is the least prime factor of n greater than 3 and $d_1 \mid n$, it follows that $d_1 = 1$ or $d_1 = 3$. If $d_1 = 1$,

then $d = 2$ and so $2^2 \equiv 1 \pmod{p}$, a contradiction. If $d_1 = 3$, then $d = 6$ and so $2^6 \equiv 1 \pmod{p}$. So $p \mid 2^6 - 1 = 63$ which in turn implies that $p = 7$. However, the order of $2 \pmod{7}$ is 3, which is odd, again a contradiction.

Therefore no such p exists, and the only solution is $n = 3$. $\square$

Example 33 (Taiwan 1997). Let $F_n = 2^{2^n} + 1$ for some positive integer n . Prove that F_n is a prime if and only if F_n is a factor of $3^{\frac{F_n-1}{2}} + 1$.

PROOF

Suppose that F_n is a factor of $3^{\frac{F_n-1}{2}} + 1$. Then $3^{\frac{F_n-1}{2}} \equiv -1 \pmod{F_n}$. By Example 8, F_n is prime.

Conversely, suppose that F_n is prime. Then, by the *Quadratic Reciprocity Law*,

$$\left(\frac{3}{F_n}\right) = (-1)^{\frac{(3-1)(F_n-1)}{4}} \left(\frac{F_n}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

By *Euler's Criterion*,

$$3^{\frac{F_n-1}{2}} \equiv \left(\frac{3}{F_n}\right) = -1 \pmod{F_n}.$$

Therefore F_n is a factor of $3^{\frac{F_n-1}{2}} + 1$. $\square$

Example 34 (China 2005 Training Problem). Let n be a positive integer and define $F_n = 2^{2^n} + 1$. Prove that for $n \geq 3$, the number F_n has a prime factor greater than $2^{n+2}(n+1)$.

PROOF

First observe that, by Example 23 with $a = 2$, if p is a prime factor of F_n , then p is of the form $2^{n+1}x_n + 1$, where x_n is a positive integer.

Next, let

$$(*) \quad F_n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k},$$

where $p_1, \dots, p_k$ are distinct primes, and $a_1, \dots, a_k$ are positive integers. Then $p_i = 2^{n+1}x_i + 1$, for some positive integers x_i with $1 \leq i \leq k$. So, on one hand, from $(*)$ we have

$$2^{2^n} + 1 \geq (2^{n+1} + 1)^{a_1 + \dots + a_k} > 2^{(n+1)(a_1 + \dots + a_k)} + 1,$$

where the last inequality follows from the *Binomial Theorem*. Consequently,

$$a_1 + \dots + a_k < \frac{2^n}{n+1}.$$

On the other hand, observe that, for $1 \leq i \leq k$, we have

$$p_i^{a_i} = (2^{n+1}x_i + 1)^{a_i} \equiv 1 + 2^{n+1}a_i x_i \pmod{2^{2n+2}}.$$

Since $n \geq 3$, it follows that $2^n \geq 2n + 2$ and so mod 2^{2n+2} , $(*)$ becomes

$$1 \equiv 2^{2^n} + 1 = \prod_{i=1}^k p_i^{a_i} \equiv \prod_{i=1}^k (1 + 2^{n+1}a_i x_i) \equiv 1 + 2^{n+1} \sum_{i=1}^k a_i x_i \pmod{2^{2n+2}}.$$

Hence $2^{n+1} \sum_{i=1}^k a_i x_i \equiv 0 \pmod{2^{2n+2}}$ which means that $\sum_{i=1}^k a_i x_i \equiv 0 \pmod{2^{n+1}}$.

Thus $\sum_{i=1}^k a_i x_i \geq 2^{n+1}$. Now let $x_j = \max\{x_1, \dots, x_k\}$. Then $x_j \sum_{i=1}^k a_i \geq 2^{n+1}$. Consequently

$$2^{n+1} \leq x_j \sum_{i=1}^k a_i < x_j \frac{2^n}{n+1}$$

and so $x_j > 2(n+1)$. Therefore $p_j = 2^{n+1}x_j + 1 > 2^{n+1}(2n+2) = 2^{n+2}(n+1)$. $\square$

Example 35. In the system of reduced residues mod p , where p is a prime, for each $e \mid p-1$, there are $\varphi(e)$ elements of order e . Prove that the sum of these $\varphi(e)$ elements is congruent to $\mu(e) \pmod{p}$, where $\mu(e)$ is the Möbius function.

PROOF

If $e \mid p-1$, let $S(e)$ denote the sum of the $\varphi(e)$ numbers of order $e \pmod{p}$ and let $T(e)$ denote the sum of the e numbers satisfying $x^e \equiv 1 \pmod{p}$. Obviously,

$$T(e) \equiv \sum_{d \mid e} S(d) \pmod{p}$$

so that, by the *Möbius Inversion Formula*,

$$S(e) \equiv \sum_{d \mid e} T(d) \mu\left(\frac{e}{d}\right) \pmod{p}.$$

But $T(e)$ is the sum of all the solutions of $P(x) = x^e - 1 \equiv 0 \pmod{p}$ in the integers mod p ; this is the negative of the coefficient of x^{e-1} in $P(x)$ so that $T(e) \equiv 0 \pmod{p}$, if $e > 1$, and $T(1) = 1$. Substitution shows that $S(e) \equiv \mu(e) \pmod{p}$. $\square$

Example 36. Let p be an odd prime, and let $S_n = 1^n + 2^n + \dots + (p-1)^n$. Prove that $S_n \equiv -1 \pmod{p}$ if $p-1 \mid n$ and $S_n \equiv 0 \pmod{p}$ otherwise.

PROOF

Let g be a primitive root mod p . Then $2, 3, \dots, p-1$ are congruent mod p , in some order, to $g, g^2, \dots, g^{p-2}$. Thus

$$(11) \quad S_n \equiv 1 + g^n + g^{2n} + \dots + g^{(p-2)n} \pmod{p}$$

If $p-1 \mid n$, let $n = (p-1)t$ for some integer t . So

$$g^{kn} = (g^{p-1})^{kt} \equiv 1 \pmod{p} \quad \text{for } k = 1, 2, \dots, p-2.$$

In this case, $S_n \equiv 1 + p - 2 \equiv -1 \pmod{p}$.

If $p-1 \nmid n$, multiply both sides of the congruence in (11) by $1 - g^n$ and get

$$(1 - g^n)S_n \equiv 1 - g^{(p-1)n} \pmod{p}.$$

Since $g^{p-1} \equiv 1 \pmod{p}$, it follows that $1 - g^{(p-1)n} \equiv 0 \pmod{p}$. But since $p-1 \nmid n$ and the order of $g \pmod{p}$ is $p-1$, it follows that $g^n \not\equiv 1 \pmod{p}$. Thus $S_n \equiv 0 \pmod{p}$. $\square$

Example 37. Let p be an odd prime, and let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{p-1}x^{p-1}$ be a polynomial of degree $p-1$ with integral coefficients. Suppose that $p \nmid (f(b) - f(a))$ whenever a and b are integers such that $p \nmid (b - a)$. Prove that $p \mid a_{p-1}$.

PROOF

Because $a \not\equiv b \pmod{p}$ implies that $f(a) \not\equiv f(b) \pmod{p}$, it follows that $f(0), f(1), \dots, f(p-1)$ form a complete residue system modulo p . Thus

$$(*) \quad 0 \equiv \sum_{i=0}^{p-1} i \equiv \sum_{j=0}^{p-1} f(j) \equiv p a_0 + \sum_{k=1}^{p-1} a_k \left(\sum_{j=0}^{p-1} j^k \right) \pmod{p}.$$

Let $S_k = \sum_{j=0}^{p-1} j^k$. To evaluate S_k , we need to consider two cases.

If $k = p-1$, then by *Fermat's Little Theorem*,

$$\sum_{j=0}^{p-1} j^k \equiv \sum_{j=0}^{p-1} 1 = p-1 \pmod{p}.$$

If $1 \leq k \leq p-2$, let g be a primitive root mod p . Then

$$S_k = \sum_{j=0}^{p-1} j^k \equiv \sum_{m=1}^{p-1} (g^m)^k \equiv \sum_{m=1}^{p-1} (g^k)^m \pmod{p},$$

so

$$g^k S_k \equiv \sum_{m=1}^{p-1} (g^k)^{m+1} \equiv \sum_{m=2}^p (g^k)^m \equiv \sum_{m=1}^{p-1} (g^k)^m \equiv S_k \pmod{p}.$$

Because $g^k \not\equiv 1 \pmod{p}$, it follows that $S_k \equiv 0 \pmod{p}$. Using these results in $(*)$, we find $0 \equiv (p-1)a_{p-1} \pmod{p}$, and it follows that $p \mid a_{p-1}$. $\square$

Example 38 (USA TST 2002). Let p be a prime greater than 5. For any integer x , define

$$f_p(x) = \sum_{k=1}^{p-1} \frac{1}{(xp+k)^2}.$$

Prove that for all positive integers x and y , the numerator of $f_p(x) - f_p(y)$ when written in lowest terms, is divisible by p^3 .

PROOF

We use the notation $r \equiv s \pmod{n}$, for r and s rational numbers, to mean that the numerator of $r - s$, when written in lowest terms, is divisible by n . This relation is symmetric and transitive, just like congruence for integers.

It is enough to check that $f_p(x) \equiv f_p(x+1) \pmod{p^3}$, or in other words,

$$\begin{aligned} 0 &\equiv \sum_{i=1}^{p-1} \left(\frac{1}{(xp+i)^2} - \frac{1}{(xp+p+i)^2} \right) \\ &= \sum_{i=1}^{p-1} \frac{(xp+p+i)^2 - (xp+i)^2}{(xp+i)^2(xp+p+i)^2} \\ &= \sum_{i=1}^{p-1} \frac{p(2xp+2i+p)}{(xp+i)^2(xp+p+i)^2} \pmod{p^3}. \end{aligned}$$

Of course, it suffices to show that after dividing both sides by p , the results are congruent mod p^2 . For integers y and z , $(y + zp)^2 \equiv y(y + 2zp) \pmod{p^2}$, so

$$(y + zp)^2(y - 2zp) \equiv y(y + 2zp)(y - 2zp) \equiv y^3 \pmod{p^2}.$$

Further, suppose that y is not divisible by p . It follows that

$$\frac{1}{(y + zp)^2} \equiv \frac{y - 2zp}{y^3} \pmod{p^3}.$$

Thus

$$\begin{aligned} \sum_{i=1}^{p-1} \frac{2i + p(2x + 1)}{(xp + i)^2(xp + p + i)^2} &\equiv \sum_{i=1}^{p-1} \frac{2i + p(2x + 1)}{i^6} (i - 2xp)(i - 2p(x + 1)) \\ &= \sum_{i=1}^{p-1} \frac{2i + p(2x + 1)}{i^6} (i^2 - 2p(2x + 1)i + 4p^2x(x + 1)) \\ &\equiv \sum_{i=1}^{p-1} \frac{2i + p(2x + 1)}{i^5} (i - 2p(2x + 1)) \\ &= \sum_{i=1}^{p-1} \frac{2i^2 + pi(2x + 1) - 4pi(2x + 1) - 2p^2(2x + 1)^2}{i^5} \\ &\equiv \sum_{i=1}^{p-1} \frac{2}{i^3} + 3p(2x + 1) \sum_{i=1}^{p-1} \frac{1}{i^4} \pmod{p^2}. \end{aligned}$$

For a prime $p \geq 7$, we apply Example 36 with $n = -4$. So

$$\sum_{i=1}^{p-1} \frac{2i + p(2x + 1)}{(xp + i)^2(xp + p + i)^2} \equiv \sum_{i=1}^{p-1} \frac{2}{i^3} \pmod{p^2}.$$

Now note that

$$\sum_{i=1}^{p-1} \frac{2}{i^3} \equiv \sum_{i=1}^{p-1} \left(\frac{1}{i^3} - \frac{1}{(p - i)^3} \right) \equiv \sum_{i=1}^{p-1} \frac{p(p^2 - 3pi + 3i^2)}{i^3(p - i)^3} \equiv \sum_{i=1}^{p-1} \frac{3i^2p}{i^3(p - i)^3} \equiv \sum_{i=1}^{p-1} -\frac{3p}{i^4} \equiv 0 \pmod{p^2}.$$

The result follows. $\square$

Example 39 (China 2003). The sides of a triangle have integer lengths k, m , and n . Assume that $k > m > n$ and

$$\left\{ \frac{3^k}{10^4} \right\} = \left\{ \frac{3^m}{10^4} \right\} = \left\{ \frac{3^n}{10^4} \right\}.$$

Determine the minimum value of the perimeter of the triangle.

SOLUTION

It suffices to find positive integers k, m , and n with $k > m > n$ and $k < m + n$ such that

$$3^k \equiv 3^m \equiv 3^n \pmod{10^4},$$

or equivalently,

$$(*) \quad 3^k \equiv 3^m \equiv 3^n \pmod{2^4} \quad \text{and} \quad 3^k \equiv 3^m \equiv 3^n \pmod{5^4}.$$

Let $d_1 = \text{ord}_{2^4} 3$, $d_2 = \text{ord}_{5^4} 3$, and $d = \gcd(d_1, d_2)$. By Theorem 1, d divides both $k - m$ and $m - n$.

It is easy to check that $d_1 = 4$. We note that d_2 divides $\varphi(5^4) = 5^4 - 5^3 = 500$, Theorem 1 again. We claim that $d_2 = 500$. If $d_2 < 500$, it must be a divisor of either $250 = \frac{500}{2}$ or $100 = \frac{500}{5}$. It suffices to show that

- (a) $3^{250} \not\equiv 1 \pmod{5^4}$ and
- (b) $3^{100} \not\equiv 1 \pmod{5^4}$.

We establish (a) by noting that $3^{250} \equiv 3^2 \equiv -1 \pmod{5}$, since $\varphi(5) = 4$. By the *Binomial Theorem*, we have

$$3^{100} = (10 - 1)^{50} \equiv \binom{50}{48} \cdot 10^2 - \binom{50}{49} \cdot 10 + 1 \not\equiv 1 \pmod{5^4},$$

establishing (b). It follows that $d_2 = 500$ and $d = 500$. Condition $(*)$ is satisfied if and only if both of $k - m$ and $m - n$ are multiples of $d = 500$.

We set $m = 500s + n$ and $k = 500t + m = 500(s + t) + n$ for positive integers s, t . The perimeter of the triangle is equal to $k + m + n = 500(2s + t) + 3n$. Condition $k < m + n$ now reads $500t < n$. Therefore, the minimum value of the perimeter is equal to $500 \cdot 3 + 3 \cdot 501 = 3003$, obtained when $s = t = 1$ and $n = 501$. $\square$

Example 40 (Turkey 2000). Find the number of ordered quadruples (x, y, z, w) of integers with $0 \leq x, y, z, w \leq 36$ such that $x^2 + y^2 \equiv z^3 + w^3 \pmod{37}$.

PROOF

For each k between 0 and 36 inclusive, we find the number of pairs (x, y) of integers with $0 \leq x, y \leq 36$ satisfying $x^2 + y^2 \equiv k \pmod{37}$. Notice that this is equivalent to $(x + 6y)(x - 6y) \equiv k \pmod{37}$.

First we consider the case $k = 0$. For each $y \in \{0, 1, \dots, 36\}$, we have $(x + 6y)(x - 6y) \equiv 0 \pmod{37}$ if and only if $x \equiv \pm 6y \pmod{37}$. Thus, there is one pair (x, y) with $y = 0$ such that $x^2 + y^2 \equiv 0 \pmod{37}$ (namely, $(x, y) = (0, 0)$), and for any other y there are two such pairs (x, y) . Hence, there are a total of $2 \cdot 36 + 1 = 73$ pairs such that $x^2 + y^2 \equiv 0 \pmod{37}$.

Now we consider the case when $k \neq 0$. Let $a \equiv x + 6y \pmod{37}$, $b \equiv x - 6y \pmod{37}$. For any value $a \in \{1, 2, \dots, 36\}$, there is exactly one value $b \in \{1, 2, \dots, 36\}$ such that $ab \equiv k \pmod{37}$. For each of these 36 pairs (a, b) corresponds to a unique solution (x, y) , because we must have $x \equiv (a + b)2^{-1} \pmod{37}$, $y \equiv (a - b)12^{-1} \pmod{37}$. Thus, the equation $(x + 6y)(x - 6y) \equiv k \pmod{37}$ has exactly 36 solutions (x, y) whenever $k \not\equiv 0 \pmod{37}$.

We proceed to count the number of ordered quadruples (x, y, z, w) of integers with $0 \leq x, y, z, w \leq 36$ such that $x^2 + y^2 \equiv z^3 + w^3 \pmod{37}$. Now observe that there are three cube roots r_1, r_2, r_3 of 1 mod 37 — namely, if we let g be a primitive root mod 37, then the cube roots are $1, g^{12}, g^{24}$. Given any z , we have $z^3 + w^3 \equiv 0 \pmod{37}$ if and only if $w = -r_1z, -r_2z$ or $-r_3z$. Hence there are 109 pairs (z, w) such that $z^3 + w^3 \equiv 0 \pmod{37}$ — one pair such that $z = 0$, and 3 pairs such that $z = z_0$ for each $z_0 \in \{1, 2, \dots, 36\}$. Above, we found that there are exactly 73 pairs (x, y) such that $x^2 + y^2 \equiv 0 \pmod{37}$. Therefore, there are $109 \cdot 73$ quadruples such that $x^2 + y^2 \equiv z^3 + w^3 \equiv 0 \pmod{37}$.

For each of the $37^2 - 109$ pairs such that $z^3 + w^3 \not\equiv 0 \pmod{37}$, there are exactly 36 pairs (x, y) such that $x^2 + y^2 \equiv z^3 + w^3 \pmod{37}$. Hence there $(37^2 - 109) \cdot 36$ quadruples (x, y, z, w) such that $x^2 + y^2 \equiv z^3 + w^3 \not\equiv 0 \pmod{37}$.

Therefore there are $109 \cdot 73 + (37^2 - 109) \cdot 36 = 53317$ quadruples (x, y, z, w) such that $x^2 + y^2 \equiv z^3 + w^3 \equiv 0 \pmod{37}$. $\square$

Example 41 (Canada 1989 Training Problem). Let m, n be positive integers. Find the minimum value of $|12^m - 5^n|$.

SOLUTION

The minimum value is $7 = 12^1 - 5^1$.

First observe that $|12^m - 5^n|$ is an odd number. Since $|12^m - 5^n|$ is neither divisible by 3 nor by 5, it follows that $|12^m - 5^n| \neq 3, 5$.

If $12^m - 5^n = 1$, then take mod 4 and get $-1 \equiv 1 \pmod{4}$, which is obviously impossible. If $5^n - 12^m = 1$, then $n > 0$. Take mod 3 and get $(-1)^n \equiv 1 \pmod{3}$. So n must be even, say $n = 2k$ for some positive integer k . Since $2^{2+m} \equiv -2^m \equiv 5^n - 12^m \equiv 1 \pmod{5}$ and $\text{ord}_5 2 = 4$, it follows that $2 + m \equiv 0 \pmod{4}$ or equivalently, $m \equiv -2 \equiv 2 \pmod{4}$. Now let $m = 4\ell + 2$ for some integer ℓ . Then

$$5^n - 12^m = (5^k)^2 - (12^{2\ell+1})^2 = (5^k + 12^{2\ell+1})(5^k - 12^{2\ell+1}) \geq 5^k + 12^{2\ell+1} > 1$$

a contradiction. Therefore $|12^m - 5^n| \geq 7$. $\square$

Example 42 (Bulgaria 1996). Find all prime numbers p, q for which pq divides $(5^p - 2^p)(5^q - 2^q)$.

SOLUTION

If $p \mid 5^p - 2^p$, then $p \mid 5 - 2$ by the *Fermat's Little Theorem*, so $p = 3$. Suppose $p, q \neq 3$. Then $q \mid 5^p - 2^p$ and $p \mid 5^q - 2^q$. Without loss of generality, assume that $p > q$, so that $\gcd(p, q - 1) = 1$. If a is an integer such that $2a \equiv 5 \pmod{q}$, then $\text{ord}_q a \mid p$ (because $5^p \equiv 2^p \pmod{q}$ and $2^p a^p \equiv 5^p \equiv 2^p \pmod{q}$ imply that $a^p \equiv 1 \pmod{q}$) as well as $\text{ord}_q a \mid q - 1$. This is clearly impossible (because $\gcd(p, q - 1) = 1$ and $p, q \geq 3$).

Hence one of p, q is equal to 3. If $q \neq 3$, then $q \mid 5^3 - 2^3 = 9 \cdot 13$, so $q = 13$. Similarly $p \in \{3, 13\}$. Therefore the solutions are $(p, q) = (3, 3), (3, 13), (13, 3)$. $\square$

Example 43 (Romania 1996). Find all prime numbers p, q for which the congruence

$$n^{3pq} \equiv n \pmod{3pq}$$

holds for all integers n .

SOLUTION

Without loss of generality, assume that $p \leq q$. If $n^{3pq} \equiv n \pmod{3pq}$ for all integers n then $n^{3pq} \equiv n \pmod{3}$ for all integers n . In particular, we have $2^{3pq} \equiv 2 \pmod{3}$ and so $2^{3pq-1} \equiv 1 \pmod{3}$. Since $\text{ord}_3 2 = 2$, we have $2 \mid 3pq - 1$ which means that p and q are odd. Moreover, if

n is a primitive root mod p , then $n^{3pq-1} \equiv 1 \pmod{p}$ implies that $p-1$ divides $3pq-1$ as well as $3pq-1-3q(p-1)=3q-1$. Similarly, $q-1$ divides $3p-1$. If $p=q$, then since

$$\frac{3p-1}{p-1} = 3 + \frac{2}{p-1}$$

it follows that $p=q=3$. But $4^{27} \equiv 1 \pmod{27}$. Hence $p < q$.

Since p and q are odd primes, $p+2 \leq q$. Hence $p+1 \leq q-1$ which in turn implies that

$$\frac{3p-1}{q-1} \leq \frac{3p-1}{p+1} = 3 - \frac{4}{p+1} < 3.$$

Because $\frac{3p-1}{q-1}$ is an integer which is greater than 1, it must be equal to 2, that is, $2q = 3p + 1$.

On the other hand, $p-1$ divides $3q-1 = \frac{9p+1}{2}$ as well as $(9p+1) - 9(p-1) = 10$. Hence $p = 11$, $q = 17$. Therefore the prime numbers p and q are 11 and 17. These solutions can be checked by using the *Fermat's Little Theorem* and *Chinese Remainder Theorem*. $\square$

Example 44. Find all nonnegative integers x, y such that $1 + 3^x = 2^y$.

SOLUTION

If $x > 1$, then $2^y \equiv 1 \pmod{9}$. Since $\text{ord}_9 2 = 6$, we have $6 \mid y$, say $y = 6r$ for some positive integer r . But $\text{ord}_7 2 = 3$. So we have $2^y \equiv 2^{6r} = (2^3)^{2r} \equiv 1 \pmod{7}$. Thus 7 divides $2^y - 1 = 3^x$, which is obviously impossible. Consequently $x \leq 1$. Therefore the solutions are $(x, y) = (0, 1), (1, 2)$. $\square$

Example 45. Find all nonnegative integers x, y such that $1 + 2^x = 3^y$.

SOLUTION

If $x > 3$, then $3^y \equiv 1 \pmod{16}$. Since $\text{ord}_{16} 3 = 4$, it follows that $y \equiv 0 \pmod{4}$. This, together with $\text{ord}_5 3 = 4$, yields $5 \mid 3^y - 1 = 2^x$ which is clearly impossible. Therefore $x \leq 3$. Consequently the solutions are $(x, y) = (3, 2), (1, 1)$. $\square$

Example 46 (Proposed by Hong Kong for IMO 1991). Find all positive integers x, y, z such that $3^x + 4^y = 5^z$.

SOLUTION

Take mod 3 and get $1 \equiv 5^z \equiv 2^z \pmod{3}$. Hence z is even because $\text{ord}_3 2 = 2$. Let $z = 2z_1$ for some positive integer z_1 . Then the equation becomes $(5^{z_1} - 2^y)(5^{z_1} + 2^y) = 3^x$. Each factor $5^{z_1} - 2^y$ and $5^{z_1} + 2^y$ is a power of 3 and $\gcd(5^{z_1} - 2^y, 5^{z_1} + 2^y) = 1$, we see that the only possibility is $5^{z_1} + 2^y = 3^x$ and $5^{z_1} - 2^y = 1$. Take mod 3 again and get $(-1)^{z_1} + (-1)^y \equiv 0 \pmod{3}$ and $(-1)^{z_1} - (-1)^y \equiv 1 \pmod{3}$. These two equations imply that z_1 is odd and y is even. Hence $y \geq 2$. For the equation $5^{z_1} + 2^y = 3^x$ we reduce it mod 4 to get $1 \equiv (-1)^x \pmod{4}$. So x is also even. Now if $y \geq 4$, we reduce the equation $5^{z_1} - 2^y = 1 \pmod{8}$ to get $5 \equiv 1 \pmod{8}$, which is impossible. Consequently, $y = 2$, $z_1 = 1$, $x = 2$.

Therefore the only solution is $(x, y, z) = (2, 2, 2)$. $\square$

Example 47. Find all nonnegative integers x, y, z such that $1 + 3^x = 2^y 5^z$.

SOLUTION

If x is odd, then $2^y 5^z \equiv 1 + 3 \pmod{8}$ and so $y = 2$. Also, if x is odd, then $1 + 3^x \not\equiv 0 \pmod{5}$, so $z = 0$. Then $1 + 3^x = 4$ gives the only solution with x odd, namely $(x, y, z) = (1, 2, 0)$.

If x is even, then $2^y 5^z \equiv 1 + 1 \pmod{4}$ and so $y = 1$. If $z > 1$, then $1 + 3^x \equiv 0 \pmod{25}$ which in turn gives $3^{x-1} \equiv 8 \pmod{25}$. Hence $3^{x+10} \equiv 8 \cdot 3^{11} \equiv 1 \pmod{25}$. Since $\text{ord}_{25} 3 = 20$, it follows that $x \equiv -10 \equiv 10 \pmod{20}$. Thus $1 + 3^x = 1 + (3^{10})^{1+2t}$ for some integer t . Now observe that the prime 1181 divides $1 + 3^{10}$. Hence $2^y 5^z = 1 + 3^x \equiv 1 + (-1)^{1+2t} \equiv 0 \pmod{1181}$, which is impossible. So $z \leq 1$. The only solution with x even is $(x, y, z) = (2, 1, 1)$.

Put the two cases together and get $(x, y, z) = (1, 2, 0), (2, 1, 1)$. □

Example 48 (Bulgaria 2006). Find all positive integers t, x, y, z such that

$$(*) \quad 2^t = 3^x 5^y + 7^z.$$

SOLUTION

First note that $2^t \equiv 1 \pmod{3}$. So t must be even because the order of $\text{ord}_3 2 = 2$. Also $2^t \equiv 2^z \pmod{5}$, or equivalently, $2^{t-z} \equiv 1 \pmod{5}$ (where obviously $t > z$). Then $4 \mid t - z$ because $\text{ord}_5 2 = 4$. Hence $2 \mid z$. Furthermore, it is clear that $t \geq 6 > 2$. So $0 \equiv 3^x (-3)^y + (-1)^z \pmod{8}$, or equivalently, $3^{x+y} \equiv (-1)^{y+1} \pmod{8}$. If y is even, then $3^{x+y} \equiv -1 \pmod{8}$ which is impossible. So y must be odd and $3^{x+y} \equiv 1 \pmod{8}$. Hence $x + y$ is even because $\text{ord}_8 3 = 2$. Now let $t = 2m$ with $m \geq 3$ and let $z = 2n$ with $n \geq 1$. Then rewrite $(*)$ as

$$(2^m - 7^n)(2^m + 7^n) = 3^x 5^y.$$

Since $\gcd(2^m - 7^n, 2^m + 7^n) = 1$, it follows that there are three possible cases to consider:

- $2^m - 7^n = 3^x, 2^m + 7^n = 5^y$;
- $2^m - 7^n = 5^y, 2^m + 7^n = 3^x$;
- $2^m - 7^n = 1, 2^m + 7^n = 3^x 5^y$.

In the first two cases we get $2^m \mp 7^n = 3^x$. Noting that $m \geq 3$ and x is odd, we obtain $\mp(-1)^n \equiv 3 \pmod{8}$, or equivalently, $3 \equiv \pm 1 \pmod{8}$, which is impossible.

In the third case, observe that $2^m \equiv 1 \pmod{7}$. Hence $3 \mid m$ (because the order of 2 mod 7 is 3). Now let $m = 3k$ for some positive integer k . Then $(2^k - 1)(2^{2k} + 2^k + 1) = 7^n$. It is easy to see that $\gcd(2^k - 1, 2^{2k} + 2^k + 1)$ is either 1 or 3. Hence $2^k - 1 = 1, 2^{2k} + 2^k + 1 = 7^n$. Consequently, $k = 1, n = 1, m = 3, t = 6, z = 2$. So $x = y = 1$.

Therefore the solution is $(t, x, y, z) = (6, 1, 1, 2)$. □

Example 49 (China 2005). Find all nonnegative integers x, y, z, w such that

$$2^x 3^y - 5^z 7^w = 1.$$

SOLUTION

Since $5^z 7^w + 1$ is even, it follows that $x \geq 1$.

- **CASE 1:** If $y = 0$, then $2^x - 5^z 7^w = 1$.

If $z \neq 0$, then $2^x \equiv 1 \pmod{5}$ and so $4 \mid x$ because the $\text{ord}_5 2 = 4$. Hence $3 \mid 2^x - 1 = 5^z 7^w$, which is obviously impossible.

If $z = 0$, then $2^x - 7^w = 1$. Test $x = 1, 2, 3$ and get $(x, w) = (1, 0), (3, 1)$ as solutions.

If $x \geq 4$, then $7^w \equiv -1 \pmod{16}$, which is impossible because when w is even this gives $1 \equiv -1 \pmod{16}$ and when w is odd this gives $7 \equiv -1 \pmod{16}$.

Therefore, if $y = 0$, the solutions are $(x, y, z, w) = (1, 0, 0, 0), (3, 0, 0, 1)$.

- **CASE 2:** If $y > 0$, $x = 1$, then $2 \cdot 3^y - 5^z 7^w = 1$. So $-5^z 7^w \equiv 1 \pmod{3}$. Hence $-(-1)^z \equiv 1 \pmod{3}$ and so z must be odd. Thus $2 \cdot 3^y \equiv 1 \pmod{5}$ which in turn implies that $3^y \equiv 2 \cdot 3 \cdot 3^y \equiv 3 \pmod{5}$. So $3^{y-1} \equiv 1 \pmod{5}$. Since $\text{ord}_5 3 = 4$, it follows that $y \equiv 1 \pmod{4}$. If $w \neq 0$, then $2 \cdot 3^y \equiv 1 \pmod{7}$ which in turn implies that $3^y \equiv 4 \pmod{7}$. So $3^{y+2} \equiv 36 \equiv 1 \pmod{7}$. Since $\text{ord}_7 3 = 6$, it follows that $y \equiv -2 \equiv 4 \pmod{6}$. This contradicts the fact that $y \equiv 1 \pmod{4}$ (because one is even and the other is odd). So $w = 0$ and $2 \cdot 3^y - 5^z = 1$. If $y = 1$, then $z = 1$. If $y \geq 2$, then $5^z \equiv -1 \pmod{9}$ and so $5^{z+3} \equiv 8 \cdot 125 \equiv 1 \pmod{9}$. Since $\text{ord}_9 5 = 6$, it follows that $z \equiv -3 \equiv 3 \pmod{6}$. Hence $5^3 + 1 \mid 5^z + 1$ implies that $7 \mid 5^z + 1 = 2 \cdot 3^y$ which is impossible.

Therefore, if $y > 0$, $x = 1$, the solution is $(x, y, z, w) = (1, 1, 1, 0)$.

- **CASE 3:** If $y > 0$, $x \geq 2$, then $5^z 7^w \equiv -1 \pmod{4}$ and $5^z 7^w \equiv -1 \pmod{3}$. Hence $(-1)^w \equiv -1 \pmod{4}$ and $(-1)^z \equiv -1 \pmod{3}$. So w and z are odd. Consequently, $2^x 3^y = 5^z 7^w + 1 \equiv 35 + 1 \equiv 4 \pmod{8}$. Thus $x = 2$ (otherwise $0 \equiv 4 \pmod{8}$) and so $4 \cdot 3^y - 5^z 7^w = 1$ (with w, z odd). Hence $4 \cdot 3^y \equiv 1 \pmod{5}$ and $4 \cdot 3^y \equiv 1 \pmod{7}$. Now note again that $\text{ord}_5 3 = 4$ and $\text{ord}_7 3 = 6$. So $y \equiv 2 \pmod{4}$ and $y \equiv 2 \pmod{6}$. Consequently, $y = 12m + 2$ for some nonnegative integer m . Then

$$5^z 7^w = 4 \cdot 3^y - 1 = (2 \cdot 3^{6m+1} - 1)(2 \cdot 3^{6m+1} + 1).$$

Since $2 \cdot 3^{6m+1} + 1 \equiv 6 \cdot 2^{3m} + 1 \equiv 6 + 1 \equiv 0 \pmod{7}$, and $\gcd(2 \cdot 3^{6m+1} - 1, 2 \cdot 3^{6m+1} + 1) = 1$, it follows that $5 \mid 2 \cdot 3^{6m+1} - 1$. Hence

$$2 \cdot 3^{6m+1} - 1 = 5^z, \quad 2 \cdot 3^{6m+1} + 1 = 7^w.$$

If $m \geq 1$, then $5^z \equiv -1 \pmod{9}$. From CASE 2 this is impossible. So $m = 0$ and get $y = 2$, $z = 1$ and $w = 1$. Therefore, if $y > 0$, $x \geq 2$, the solution is $(x, y, z, w) = (2, 2, 1, 1)$.

Put all the cases together and get the solutions $(x, y, z, w) = (1, 0, 0, 0), (3, 0, 0, 1), (1, 1, 1, 0), (2, 2, 1, 1)$. □

Example 50 (Russia 2000). Determine whether there exist pairwise relatively prime integers a, b and c with $a, b, c > 1$ such that $b \mid 2^a + 1$, $c \mid 2^b + 1$, $a \mid 2^c + 1$.

PROOF

No such integers exist.

First assume that all a, b, c are primes. By cyclic conditions given in the problem, we may assume that $a < b$ and $a < c$. By *Fermat's Little Theorem* and Theorem 1, $\text{ord}_a 2$ divides $\gcd(2c, a-1) = 2$, by noting that c is a prime greater than a . Since a is an odd prime, $\text{ord}_a 2$ must then be 2, implying that $a = 3$, and so $b \mid 2^a + 1 = 9$, which is a contradiction.

What if a, b, c are not all primes? We try to generalize our previous method. Let $\pi(n)$ denote the smallest prime factor of a positive integer n . We make the following claim:

If p is a prime such that $p \mid 2^y + 1$ and $p < \pi(n)$, then $p = 3$.

The proof of our claim is similar to our previous discussion for the case that all a, b, c are primes. Then $\text{ord}_p 2$ divides $\gcd(2y, p-1) = 2$. Again $\text{ord}_p 2 = 2$ and $p = 3$, establishing our claim.

Now we solve our main problem. Since a, b, c are pairwise relatively prime, $\pi(a), \pi(b)$ and $\pi(c)$ are distinct. Without loss of generality, assume that $\pi(a) < \pi(b), \pi(c)$. Applying the claim with $(p, y) = (\pi(a), c)$, we find that $\pi(a) = 3$. Write $a = 3a_0$. We claim that $3 \nmid a_0$. Otherwise, 9 would divide $2^c + 1$ and hence $2^{2c} - 1$. Because $2^n \equiv 1 \pmod{9}$ only if $6 \mid n$, we must have $6 \mid 2c$. Then $3 \mid c$ contradicting the assumption that a and c are relatively prime. Thus 3 does not divide a_0, b, c . Let $q = \pi(a_0bc)$ so that $\pi(q) = q \leq \min\{\pi(b), \pi(c)\}$.

Suppose, for the sake of contradiction, that $q \mid a$. Because a and c are relatively prime, $q \nmid c$, implying that $\pi(q) = q$ is not equal to $\pi(c)$. Because $\pi(q) \leq \pi(c)$, we must have $\pi(q) < \pi(c)$. Furthermore, q must divide $2^c + 1$ because it divides a factor of $2^c + 1$ (namely a). Applying our claim to $(p, y) = (q, c)$, we find that $q = 3$, a contradiction. Hence, our assumption was wrong, and $q \nmid a$. Similarly, $q \nmid c$. It follows that q must divide b .

Now let $\text{ord}_q 2 = e$. Then $e \leq q-1$, so the prime factors of e are less than q . Also, $q \mid b$ and hence $q \mid 2^a + 1$ and $q \mid 2^{2a} - 1$, implying that $e \mid 2a$. The only prime factors of $2a$ less than q are 2 and 3, so $e \mid 6$. Thus $q \mid 2^6 - 1$, and $q = 7$. However, $2^3 \equiv 1 \pmod{7}$, so

$$2^a + 1 \equiv (2^3)^{a_0} + 1 \equiv 1^{a_0} + 1 = 2 \pmod{7}.$$

Hence, $q \nmid 2^a + 1$, contradicting the assumption that q divides b which divides $2^a + 1$. $\square$

Example 51 (USA TST 2003 Reid Barton). Find all ordered triples of primes (p, q, r) such that $p \mid q^r + 1$, $q \mid r^p + 1$, $r \mid p^q + 1$.

SOLUTION

The answers are $(2, 5, 3)$ and its cyclic permutations.

We check that this is a solution:

$$2 \mid 126 = 5^3 + 1, \quad 5 \mid 10 = 3^2 + 1, \quad 3 \mid 33 = 2^5 + 1.$$

Now let p, q, r be three primes satisfying the given divisibility relations. Since $q \nmid q^r + 1$, $p \neq q$, and similarly $q \neq r$, $r \neq p$, so p, q, r are all distinct.

We first consider the case that p, q, r are all odd. Since $p \mid q^r + 1$, by Example 22, either $2r \mid p - 1$ or $p \mid q^2 - 1$. But $2r \mid p - 1$ is impossible because $2r \mid p - 1$ leads to $p \equiv 1 \pmod{r}$, or equivalently, $0 \equiv p^q + 1 \equiv 1^q + 1 = 2 \pmod{r}$, which contradicts the fact that $r > 2$. Thus we must have $p \mid q^2 - 1 = (q - 1)(q + 1)$. Since p is an odd prime and $q - 1, q + 1$ are both even, we must have $p \mid \frac{q-1}{2}$ or $p \mid \frac{q+1}{2}$; either way, $p \leq \frac{q+1}{2} < q$. But then by a similar argument we may conclude that $q < r$, $r < p$, a contradiction.

Thus, at least one of p, q, r must be 2. By a cyclic permutation we may assume that $q = 2$. Now $p \mid q^r + 1$, so by Example 22 again, either $2r \mid p - 1$ or $p \mid 2^2 - 1 = 3$. But $2r \mid p - 1$ is impossible as before, because $p^q + 1 = p^2 + 1 = (p^2 - 1) + 2$ and $r > 2$. Hence, we must have $p = 3$, and $r \mid p^q + 1 = 3^2 + 1 = 10$. Because $r \neq q$, we must have $r = 5$. Therefore $(2, 5, 3)$ and its cyclic permutations are the only solutions. $\square$

Example 52 (Turkey 1998). Let $f(x_1, \dots, x_n)$ be a polynomial with integer coefficients of total degree less than n . Show that the number of ordered n -tuples $(x_1, \dots, x_n)$ with $0 \leq x_i \leq 12$ such that $f(x_1, \dots, x_n) \equiv 0 \pmod{13}$ is divisible by 13.

PROOF

We claim that, for $0 \leq k < 12$,

$$\sum_{x=0}^{12} x^k \equiv 0 \pmod{13}.$$

The case $k = 0$ is obvious, so let us assume $k > 0$. First, observe that 2 is a primitive root mod 13, meaning that 2^m , $m \geq 1$, exhausts all nonzero residues mod 13. so, on the one hand, $2^k \not\equiv 1 \pmod{13}$ for $1 \leq k < 12$, and on the other hand, the residue classes $2, 4, 6, \dots, 24$ are a permutation of the residue classes $1, 2, \dots, 12$. We deduce that

$$\sum_{x=0}^{12} x^k \equiv \sum_{x=0}^{12} (2x)^k = 2^k \sum_{x=0}^{12} x^k \pmod{13},$$

and because $2^k \not\equiv 1 \pmod{13}$, we must have $\sum_{x=0}^{12} x^k \equiv 0 \pmod{13}$.

Now let $S = \{(x_1, \dots, x_n) : 0 \leq x_i \leq 12\}$. Because $|S| = 13^n$ is divisible by 13, it suffices to show that the number of ordered n -tuples $(x_1, \dots, x_n) \in S$ such that $f(x_1, \dots, x_n) \not\equiv 0 \pmod{13}$ is divisible by 13. Consider the sum

$$\sum_{(x_1, \dots, x_n) \in S} (f(x_1, \dots, x_n))^{12}.$$

This sum is congruent mod 13 to the number of ordered n -tuples $(x_1, \dots, x_n) \in S$ such that $f(x_1, \dots, x_n) \not\equiv 0 \pmod{13}$, since by *Fermat's Little Theorem*,

$$(f(x_1, \dots, x_n))^{12} = \begin{cases} 1 & \text{if } f(x_1, \dots, x_n) \not\equiv 0 \pmod{13}, \\ 0 & \text{if } f(x_1, \dots, x_n) \equiv 0 \pmod{13}. \end{cases}$$

On the other hand, $(f(x_1, \dots, x_n))^{12}$ can be expanded as

$$(f(x_1, \dots, x_n))^{12} = \sum_{j=1}^N c_j \prod_{i=1}^n x_i^{e_{ji}}$$

for some integers N, c_j, e_{ji} . Since f is a polynomial of total degree less than n , $e_{j1} + e_{j2} + \dots + e_{jn} < 12n$ for every j , so for each j there exists an i such that $e_{ji} < 12$. Thus, by our claim

$$\sum_{(x_1, \dots, x_n) \in S} c_j \prod_{i=1}^n x_i^{e_{ji}} = c_j \prod_{i=1}^n \sum_{x_i=0}^{12} x_i^{e_{ji}} \equiv 0 \pmod{13}$$

since one of the sums in the product is 0. Therefore

$$\sum_{(x_1, \dots, x_n) \in S} (f(x_1, \dots, x_n))^{12} = \sum_{(x_1, \dots, x_n) \in S} \sum_{j=1}^N c_j \prod_{i=1}^n x_i^{e_{ji}} \equiv 0 \pmod{13}.$$

This implies that the number of ordered n -tuples $(x_1, \dots, x_n) \in S$ such that $f(x_1, \dots, x_n) \not\equiv 0 \pmod{13}$ is divisible by 13, and we are done. $\square$

Example 53 (Taiwan 2000). Suppose that $\varphi(5^m - 1) = 5^n - 1$ for some positive integers m, n . Prove that $\gcd(m, n) > 1$.

PROOF

Suppose that $\gcd(m, n) = 1$.

We first show that m is odd. Assume that m is not odd. Now note that 5^x is congruent to 1 (respectively, 5) mod 8 if x is even (respectively, odd). If m is even, then $5^m - 1$ is divisible by (but not equal to) 8. Because $5^m - 1$ does not equal 8, $5^n - 1 = \varphi(5^m - 1)$ is either divisible by $\varphi(16) = 8$ or $\varphi(8)\varphi(p^\alpha) = 4\varphi(p^\alpha) \equiv 0 \pmod{8}$ for some odd prime power $p^\alpha > 1$. Therefore n must be even, contrary to the assumption that $\gcd(m, n) = 1$.

Next suppose, for the sake of contradiction, that $p^2 \mid 5^m - 1$ for some odd prime p . Clearly, $p \nmid 5$, so 5 has some order mod p ; let d be this order. Because $p \mid 5^m - 1$, we have $d \mid m$. Also p divides $\varphi(p^2)$ and hence $p \mid \varphi(5^m - 1) = 5^n - 1$, implying that $d \mid n$ as well. But $d > 1$ because $5 \not\equiv 1 \pmod{p}$, implying that $\gcd(m, n) \neq 1$, contradiction.

Therefore,

$$5^m - 1 = 4 \prod_{p \in S} p$$

for some set S of odd primes. Let p be any element of S . Because

$$1 = \left(\frac{1}{p}\right) = \left(\frac{5^m}{p}\right) = \left(\frac{5}{p}\right)^m$$

and m is odd, $\left(\frac{5}{p}\right) = 1$. Also, by the *Quadratic Reciprocity Law*,

$$\left(\frac{5}{p}\right) \left(\frac{p}{5}\right) = (-1)^{\frac{(5-1)(p-1)}{4}} = 1,$$

implying that $\left(\frac{p}{5}\right) = 1$. Hence, p is congruent to 1 or 4 mod 5. However, we cannot have $p \not\equiv 1 \pmod{5}$ because then 5 would have to divide $p - 1 = \varphi(p)$ and hence $5 \mid \varphi(5^m - 1) = 5^n - 1$ which is impossible. Therefore $p \equiv 4 \pmod{5}$.

Consequently,

$$-1 \equiv 5^m - 1 = 4 \prod_{p \in S} p \equiv 4 \cdot 4^{|S|} \pmod{5}$$

and

$$-1 \equiv 5^n - 1 = \varphi(4) \prod_{p \in S} \varphi(p) = 2 \prod_{p \in S} (p - 1) \equiv 2 \cdot 3^{|S|} \pmod{5}.$$

From the first of these equations we have that $|S|$ must be even, but from the second we have that $|S| \equiv 3 \pmod{4}$, a contradiction. Therefore our original assumption was false, and so $\gcd(m, n) > 1$. $\square$

Example 54 (Bulgaria 1998). Let m, n be positive integers such that

$$A = \frac{(m+3)^n + 1}{3m} \text{ is an integer.}$$

Prove that A is odd.

PROOF

If m is odd, then $(m+3)^n + 1$ is odd and so A is odd.

Now suppose m is even. Since A is an integer,

$$0 \equiv (m+3)^n + 1 \equiv m^n + 1 \pmod{3},$$

so n is odd, say, $n = 2k + 1$, and $m \equiv -1 \pmod{3}$.

We now consider the following cases:

- CASE 1: $m = 8m'$ for some positive integer m' . Then $(m+3)^n + 1 \equiv 3^{2k+1} + 1 \equiv 4 \pmod{8}$ and $3m \equiv 0 \pmod{8}$. So in this case, A is not an integer.
- CASE 2: $m = 2m'$ for some odd positive integer m' , that is, $m \equiv 2 \pmod{4}$. Then $(m+3)^n + 1 \equiv (2+3)^n + 1 \equiv 2 \pmod{4}$ and $3m \equiv 2 \pmod{4}$. So in this case, A is odd.
- CASE 3: $m = 4m'$ for some odd positive integer m' . Since $m \equiv -1 \pmod{3}$, there exists an odd prime p such that $p \equiv -1 \pmod{3}$ and $p \mid m$. Since A is an integer, $0 \equiv (m+3)^n + 1 \equiv 3^{2k+1} + 1 \pmod{m}$ and $3^{2k+1} \equiv -1 \pmod{p}$. Let g be a primitive root of p and let b be a positive integer such that $g^b \equiv 3 \pmod{p}$. Then $g^{(2k+1)b} \equiv -1 \pmod{p}$. Now note that $\left(\frac{p}{3}\right) = \left(\frac{-1}{3}\right) = -1$.
 - (a) If $p \equiv 1 \pmod{4}$, then $\left(\frac{-1}{p}\right) = 1$ and so $g^{2c} \equiv -1 \equiv g^{(2k+1)b} \pmod{p}$ for some positive integer c . Hence b is even and $\left(\frac{3}{p}\right) = 1$. Again, by the *Quadratic Reciprocity Law*,

$$-1 = \left(\frac{3}{p}\right) \left(\frac{p}{3}\right) = (-1)^{\frac{(3-1)(p-1)}{4}} = 1,$$

a contradiction.

(b) If $p \equiv 3 \pmod{4}$, then $\left(\frac{-1}{p}\right) = -1$ and so $g^{2c+1} \equiv -1 \equiv g^{(2k+1)b} \pmod{p}$ for some positive integer c . Hence b is odd and $\left(\frac{3}{p}\right) = -1$. Again, by the *Quadratic Reciprocity Law*,

$$1 = \left(\frac{3}{p}\right)\left(\frac{p}{3}\right) = (-1)^{\frac{(3-1)(p-1)}{4}} = -1,$$

a contradiction.

Thus for $m = 4m'$ and m' is odd, A is not an integer.

Therefore if A is an integer, then A is odd. □

Example 55 (Putnam 1994). For any integer a , set $n_a = 101a - 100 \cdot 2^a$. Show that for $0 \leq a, b, c, d \leq 99$, $n_a + n_b \equiv n_c + n_d \pmod{10100}$ implies $\{a, b\} = \{c, d\}$.

PROOF

By the *Chinese Remainder Theorem*, $n_a + n_b \equiv n_c + n_d \pmod{10100}$ is equivalent to

$$(*) \quad a + b \equiv c + d \pmod{100}$$

and

$$(**) \quad 2^a + 2^b \equiv 2^c + 2^d \pmod{101}.$$

By *Fermat's Little Theorem*, $(*)$ implies $2^{a+b} \equiv 2^{c+d} \pmod{101}$, or equivalently

$$(\dagger) \quad 2^a 2^b \equiv 2^c 2^d \pmod{101}.$$

Solve for 2^b in $(**)$ and substitute in $(\dagger)$ to obtain

$$2^a(2^c + 2^d - 2^a) \equiv 2^c 2^d \pmod{101},$$

or equivalently,

$$0 \equiv (2^a - 2^c)(2^a - 2^d) \pmod{101}.$$

Hence $2^a \equiv 2^c \pmod{101}$ or $2^a \equiv 2^d \pmod{101}$. By Example 10, 2 is primitive root mod 101. So, by Theorem 1, $a \equiv c \pmod{100}$ or $a \equiv d \pmod{100}$. Then, by $(*)$, b is congruent to the other. But $0 \leq a, b, c, d \leq 99$ so these congruences are equalities. □

Example 56 (Richard Stanley). Let p be an odd prime. It has been conjectured that there exists a positive integer $k \leq p-1$ which is a primitive root mod p and which is relatively prime to $p-1$. Prove this conjecture in the special case $p \equiv 1 \pmod{4}$ and $3\varphi(p-1) > p-1$.

PROOF

In fact, for $p \equiv 1 \pmod{4}$ there are at least $(3\varphi(p-1) - (p-1))/2$ such k .

Let $S = \{1, \dots, p-1\}$, let A be the subset of primitive roots mod p , and let B be the subset of numbers relatively prime to $p-1$. Define $f(n) = p-n$, and note that f is a permutation of S . For $p \equiv 1 \pmod{4}$, if g is a primitive root mod p so is $p-g$ (by Example 18). We thus have $f(A) = A$.

Because $p - 1$ is even, every element of B is odd, whence every element of $f(B)$ is even; hence $B \cap f(B) = \emptyset$.

Let $|X|$ denote the number of elements in the set X ; thus $|S| = p - 1$, $|A| = |B| = |f(B)| = \varphi(p - 1)$. By the *Principle of Inclusion-Exclusion*, the number of elements in the union of A, B , and $f(B)$ is

$$|A| + |B| + |f(B)| - |A \cap B| - |A \cap f(B)|$$

since the terms $-|B \cap f(B)|$ and $+|A \cap B \cap f(B)|$ vanish. Now

$$|A \cap f(B)| = |f(A) \cap f(B)| = |f(A \cap B)| = |A \cap B|$$

so that

$$|A \cup B \cup f(B)| = 3\varphi(p - 1) - 2|A \cap B|.$$

Since $|S| \geq |A \cup B \cup f(B)|$ we have

$$2|A \cap B| \geq 3\varphi(p - 1) - (p - 1)$$

as desired. □

Example 57. Let p be a prime greater than 3. If $3\varphi(p - 1) > p - 1$, then p has consecutive primitive roots.

PROOF

First we claim that if p is a prime greater than 3 then exactly half the primitive roots mod p are followed by quadratic nonresidues. To see why, let g be a primitive root mod p . Now $p - 1$ and $p - 2$ are clearly relatively prime and so g^{p-2} is also a primitive root mod p . Note, moreover, that $g \not\equiv g^{p-2} \pmod{p}$ because $p \neq 3$. Since g is also a quadratic nonresidue, it follows from the congruence

$$g(g^{p-2} + 1) = g^{p-1} + g \equiv 1 + g \pmod{p}$$

that $g + 1$ is a quadratic nonresidue if and only if $g^{p-2} + 1$ is a quadratic residue. Thus exactly one of the primitive roots g and g^{p-2} is followed by a quadratic nonresidue. It is easy to see that if h is a primitive root incongruent to g and g^{p-2} then h^{p-2} is also incongruent to g and g^{p-2} . The claim then follows by considering all pairs of primitive roots g and g^{p-2} .

Returning to our problem, we suppose that p is a prime greater than 3 such that $3\varphi(p - 1) > p - 1$. Now we assume that there are no consecutive primitive roots mod p . Then, by the above claim, one half of the $\varphi(p - 1)$ primitive roots are followed by quadratic nonresidues, none of which is a primitive root. Since each primitive root is a quadratic nonresidue, it follows that there are then at least

$$\frac{1}{2}\varphi(p - 1) + \varphi(p - 1) = \frac{3}{2}\varphi(p - 1)$$

quadratic nonresidues. Since there are a total of $\frac{p-1}{2}$ quadratic nonresidues,

$$\frac{3}{2}\varphi(p - 1) \leq \frac{p - 1}{2},$$

a contradiction. □

Example 58. Let p be an odd prime, and let m be a positive multiple of $p - 1$. Prove that

$$\sum_{0 \leq \ell \leq \frac{m}{p-1}} \binom{m}{(p-1)\ell} \equiv 2 + p(1+m) \pmod{p^2}.$$

PROOF

Let $m = r(p-1)$ and define

$$S_r = \sum_{0 \leq \ell \leq \frac{m}{p-1}} \binom{m}{(p-1)\ell} = \sum_{i=0}^r \binom{r(p-1)}{i(p-1)}.$$

Let g be a primitive root mod p . Choosing h to be an integer $h \equiv g^p \pmod{p^2}$, we have $h^{p-1} \equiv g^{\varphi(p^2)} \equiv 1 \pmod{p^2}$. Since

$$(h^k - 1) \sum_{j=0}^{p-2} h^{jk} = h^{k(p-1)} - 1 \equiv 0 \pmod{p^2},$$

if $h^k \not\equiv 1 \pmod{p}$, then

$$\sum_{j=0}^{p-2} h^{jk} \equiv 0 \pmod{p^2}.$$

Since $h^k \equiv g^k \pmod{p}$, this occurs whenever $p-1 \nmid k$. If $p-1 \mid k$, then

$$\sum_{j=0}^{p-2} h^{jk} \equiv \sum_{j=0}^{p-2} 1 \equiv p-1 \pmod{p^2}.$$

Hence

$$\sum_{j=0}^{p-2} (1 + h^j)^{r(p-1)} = \sum_{k=0}^{r(p-1)} \binom{r(p-1)}{k} \sum_{j=0}^{p-2} h^{jk} \equiv (p-1)S_r \pmod{p^2}.$$

If $j = (p-1)/2$, then $h^j \equiv -1 \pmod{p}$, and so $(1 + h^j)^{p-1} \equiv 0 \pmod{p^2}$. Otherwise, $h^j \not\equiv -1 \pmod{p}$, and so $(1 + h^j)^{p-1} = 1 + pa_j$ for some integer a_j , a consequence of *Fermat's Little Theorem*. Hence

$$(p-1)S_r \equiv \sum_{\substack{0 \leq j \leq p-2 \\ j \neq (p-1)/2}} (1 + pa_j)^r \equiv p-2 + pr \sum_{\substack{0 \leq j \leq p-2 \\ j \neq (p-1)/2}} a_j \pmod{p^2}.$$

Multiplying by $-(p+1)$ gives

$$S_r \equiv 2 + p + prA \pmod{p^2}$$

for some integer A that does not depend on r . As $S_1 = 2$, we have $A \equiv -1 \pmod{p}$. Also $m = r(p-1) \equiv -r \pmod{p}$, so

$$S_r \equiv 2 + p - pr \equiv 2 + p + pm \pmod{p^2}.$$

□

Example 59 (Vietnam 1997). Let $f : \mathbb{N} \cup \{0\} \rightarrow \mathbb{Z}$ be the function defined by

$$f(0) = 2, \quad f(1) = 503, \quad f(n+2) = 503f(n+1) - 1996f(n).$$

For $k \in \mathbb{N}$, let $s_1, \dots, s_k$ be integers not less than k , and let p_i be a prime factor of $f(2^{s_i})$ for $i = 1, \dots, k$. Prove that for $t = 1, \dots, k$,

$$2^t \mid p_1 + \dots + p_k \text{ if and only if } 2^t \mid k.$$

PROOF

First, by induction, $f(n) = 4^n + 499^n$ for $n \geq 0$.

Next, claim that if p is an odd prime, $m, n \in \mathbb{N}$ not divisible by p , $s \geq 0$, and $p \mid m^{2^s} + n^{2^s}$, then $p \equiv 1 \pmod{2^{s+1}}$.

To prove the claim, let g be a primitive root mod p , and take k such that $g^k \equiv mn^{-1} \pmod{p}$. As $p \mid m^{2^s} + n^{2^s}$, $m^{2^s} \equiv -n^{2^s} \pmod{p}$ and so $(mn^{-1})^{2^s} \equiv -1 \pmod{p}$. Hence $g^{2^s k} \equiv -1 \pmod{p}$. Now since $g^{\frac{p-1}{2}} \equiv -1 \pmod{p}$ it follows that

$$2^s k \equiv \frac{p-1}{2} \pmod{p-1},$$

or equivalently,

$$2^s k = \frac{p-1}{2} + j(p-1) \quad \text{for some integer } j.$$

So $2^s k = \ell \left(\frac{p-1}{2} \right)$ with ℓ odd. Thus $2^s \mid \frac{p-1}{2}$ and so $2^{s+1} \mid p-1$ and this proves the claim.

Now let $k \in \mathbb{N}$, $s_1, \dots, s_k$ integers not less than k , and p_i a prime factor of $f(2^{s_i}) = 4^{2^{s_i}} + 499^{2^{s_i}}$ for $i = 1, \dots, k$. Then p_i is not 2 or 499, so by the claim, $p_i \equiv 1 \pmod{2^{s_i+1}}$; in particular, $p_i \equiv 1 \pmod{2^k}$. Hence $p_1 + \dots + p_k \equiv k \pmod{2^k}$. Therefore $2^t \mid p_1 + \dots + p_k$ if and only if $2^t \mid k$. $\square$

Example 60. Suppose an integer $m > 6$ has a primitive root r . Let $a_j \equiv r^j \pmod{m}$, where a_j is the least positive residue. Define

$$F(x) = \sum_{j=1}^{\varphi(m)} a_j x^{\varphi(m)-j}.$$

Prove that $F(x)$ is the product of $t+1$ polynomials

$$F(x) = f_1(x)f_2(x) \cdot \dots \cdot f_t(x)g(x),$$

where every coefficient of $f_i(x)$ is 1, and the sum of the coefficients of $g(x)$ is m . Here t is the number of distinct prime factors of $\frac{1}{2}\varphi(m)$.

PROOF

Let $q = \frac{1}{2}\varphi(m)$ and define

$$F_1(x) = \sum_{k=0}^{q-1} x^k, \quad h(x) = a_1 x^q + 1 + \sum_{j=1}^{q-1} (a_{j+1} - a_j) x^{q-j}.$$

Noting that $r^q \equiv m - 1 \pmod{m}$ so that $a_i + a_{q+i} = m$, $i = 0, 1, \dots, q$, we see easily that the product $F_1(x)h(x)$ is $F(x)$.

Now note that the set $\{a_j\}$ is the set $\{a : 0 < a < m, \gcd(a, m) = 1\}$. Hence

$$\sum_{j=1}^{2q} a_j = mq$$

because there are $2q$ members of $\{a_j\}$, and for every a_j in the set there is an a_k such that $a_j + a_k = m$. So $F(1) = mq$. Moreover $F_1(1) = q$, whence $h(1)$ must equal m , that is, the sum of the coefficients of $h(x)$ is m .

Let

$$(*) \quad q = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t},$$

where $p_1, \dots, p_t$ are distinct primes, and $a_1, \dots, a_t$ are positive integers.

Then $F_1(x) = \frac{x^q - 1}{x - 1}$ admits the factors

$$\frac{x^{p_1^{a_1}} - 1}{x - 1}, \frac{x^{p_2^{a_2}} - 1}{x - 1}, \dots, \frac{x^{p_t^{a_t}} - 1}{x - 1}$$

the sum of whose coefficients are $p_1^{a_1}, p_2^{a_2}, \dots, p_t^{a_t}$ respectively. If $F_2(x)$ is the product of all factors of $F_1(x)$ not listed above, we can put

$$f_1(x) = \frac{x^{p_1^{a_1}} - 1}{x - 1}, f_2(x) = \frac{x^{p_2^{a_2}} - 1}{x - 1}, \dots, f_t(x) = \frac{x^{p_t^{a_t}} - 1}{x - 1}.$$

Then $F_1(x) = f_1(x)f_2(x) \cdots f_t(x)F_2(x)$, where

$$F_1(1) = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t} \cdot F_2(1) = q.$$

Hence $F_2(1) = 1$ and we can put $g(x) = F_2(x)h(x)$ where $g(1) = F_2(1)h(1) = m$. Thus the sum of the coefficients of $g(x)$ is m , and, from the definition, all the coefficients of $f(x)$ are ones. $\square$

Example 61. For any odd prime p , let g_p and G_p be the least positive and greatest negative primitive root of p . Prove that for any positive integer $M \geq 2$, there exist infinitely many primes p for which $M < g_p < p - M$ and $-p + M < G_p < -M$.

PROOF

First claim that for any positive integer $M \geq 2$, there exist infinitely many primes p for which

$$\left(\frac{-1}{p}\right) = \left(\frac{q_i}{p}\right) = 1,$$

where q_i , $i = 1, \dots, n$, runs through the primes less than or equal to M .

To prove the claim, suppose that the prime $p = 1 + 8q_2q_3 \cdots q_n k$ for some positive integer k and $q_1 = 2$. Then, by *Euler's Criterion*,

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = 1,$$

and

$$\left(\frac{q_1}{p}\right) = \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = 1.$$

By the *Quadratic Reciprocity Law*,

$$\left(\frac{p}{q_i}\right) \left(\frac{q_i}{p}\right) = (-1)^{\frac{(p-1)(q_i-1)}{4}} = 1.$$

But $\left(\frac{p}{q_i}\right) = \left(\frac{1}{q_i}\right) = 1$ so that $\left(\frac{q_i}{p}\right) = 1$, and p has the required properties. That there are infinitely many primes of this form is a consequence of *Dirichlet's Theorem* (on primes in arithmetic progression).

Let p be a prime satisfying the conditions of the claim. Then all integers r such that $-M \leq r + tp \leq M$, where t is an integer, are quadratic residues mod p , and therefore not primitive roots of p . Since there is a primitive root between 0 and p , it follows that $M < g_p < p - M$ and, since $G_p = -g_p$, that $-p + M < G_p < -M$. $\square$

Example 62. Prove that

$$\prod_{1 \leq x < y \leq \frac{p-1}{2}} (x^2 + y^2) \equiv (-1)^{\lfloor \frac{p+1}{8} \rfloor} \pmod{p}$$

for every $p \equiv 3 \pmod{4}$.

PROOF

Write $p = 4k + 3$ and denote let

$$P = \prod_{1 \leq x < y \leq \frac{p-1}{2}} (x^2 + y^2).$$

In the product, all sums of pairs of quadratic residues are multiplied. Let g be a primitive root mod p , and put $h = g^2$. By Theorem 9, the numbers $1, h, h^2, \dots, h^{2k}$ represent every quadratic residue once; thus we have

$$P \equiv \prod_{0 \leq i < j \leq 2k} (h^i + h^j) \pmod{p}.$$

This implies

$$(*) \quad P \cdot \prod_{0 \leq i < j \leq 2k} (h^i - h^j) \equiv \prod_{0 \leq i < j \leq 2k} (h^{2i} - h^{2j}) \pmod{p}.$$

Here each product is the value of a nonzero Vandermonde determinant $V(x_1, x_2, \dots, x_n)$. Recall that $V(x_1, x_2, \dots, x_n)$ is the determinant of an $n \times n$ matrix whose i th row is given by $(1 \ x_i \ x_i^2 \ \dots \ x_i^{n-1})$. The value of $V(x_1, x_2, \dots, x_n)$ is given by

$$V(x_1, x_2, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j).$$

Thus $(*)$ can be written as

$$P \cdot V(1, h, h^2, \dots, h^{2k}) \equiv V(1, h^2, h^4, \dots, h^{4k}) \pmod{p}.$$

Since $h^{2k+1} = g^{4k+2} \equiv 1 \pmod{p}$, the generators of the second Vandermonde determinant are congruent to

$$1, h^2, \dots, h^{2k}, h, h^3, \dots, h^{2k-1}.$$

Hence the second determinant can be obtained from the first by

$$k + (k-1) + \dots + 2 + 1 = \frac{k(k+1)}{2}$$

transpositions of rows.

Therefore

$$P \equiv (-1)^{\frac{k(k+1)}{2}} \equiv (-1)^{\lfloor \frac{k+1}{2} \rfloor} \equiv (-1)^{\lfloor \frac{p+1}{8} \rfloor} \pmod{p}.$$

□

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